

Full Length Article

Neural computation as operator-theoretic learning on measures

Míngshū Wáng 

Independent Researcher, Hong Kong, China

ARTICLE INFO

Keywords:

Measure theory
Operators
Neural networks

ABSTRACT

This paper formulates neural computation as the finite-dimensional representation of operators acting on functions and measures. Dense layers, convolutions, attention mechanisms, graph networks, neural fields, and neural operators are treated as instances of kernel-based transformations evaluated through sampling, projection, quadrature, or basis truncation. The aim is to separate the operator being learned from the particular array representation used to compute it. This separation makes architecture design an approximation problem: one must choose a measure space, function space, kernel class, topology, and discretization scheme suited to the target operator. Numerical analysis provides stability and quadrature tools; approximation theory gives representation rates; harmonic analysis identifies bases and transforms adapted to structure; signal processing supplies efficient finite representations; measure theory treats empirical data, weak convergence, transport, and distributional functionals. The result is a unified account of neural architectures as computable approximations of operators over structured domains.

1. Introduction

Neural computation is usually implemented through finite arrays. Activations are stored as vectors, weights as matrices, images as sampled grids, graphs as adjacency matrices, and token sequences as finite lists of embeddings. These arrays are computational representations of objects with additional structure. A sampled image represents a function on a grid. A point cloud represents an empirical measure. A graph signal represents a function on vertices. A token sequence represents a feature-bearing empirical measure with positional information. A neural layer acts on these represented objects through kernels, projections, nonlinear updates, and finite integration rules.

This paper develops an operator-theoretic formulation of this viewpoint. Let $(X, \mathcal{A}, \mu)$ be a measure space and let $u : X \rightarrow E$ be a feature field with values in a finite-dimensional feature space E . A large class of neural layers can be written as

$$u_{l+1}(x) = \Gamma_l \left(x, u_l(x), \int_X K_l(x, y, u_l(y)) \Psi_l(u_l(y)) d\mu_l(y) \right).$$

The kernel K_l specifies interactions across the domain, Ψ_l transforms features before aggregation, and Γ_l updates the state at x . The measure μ_l may be fixed, empirical, sampled, learned, or induced by the current state. Finite neural computation appears after this operator is evaluated through a finite representation, such as samples, quadrature nodes, basis coefficients, graph vertices, Fourier modes, or tokens.

Several strands of the literature already use parts of this language. Neural field models treat neural activity as a continuum state governed by integral or integro-differential equations (Amari, 1977). Classical approximation theory for feedforward networks studies finite-dimensional nonlinear maps and their approximation properties

(Hornik et al., 1989). Kernel mean embeddings represent probability measures by elements of reproducing kernel Hilbert spaces and provide a rigorous way to compare distributions through test functions (Muan-det et al., 2017). Deep Sets characterize permutation-invariant functions through empirical aggregation and pointwise feature maps (Zaheer et al., 2017). Mean-field analyses of wide neural networks describe training and representation in terms of probability measures over parameters (Mei et al., 2018). Neural ordinary differential equations interpret depth as time and residual updates as discretized flows (Chen et al., 2018). Neural operators, including DeepONet and Fourier neural operators, learn maps between function spaces and separate the target operator from a particular discretization (Li et al., 2021; Lu et al., 2021). Scattering transforms and harmonic architectures use structured transforms, multiscale representations, and group symmetries to construct stable features (Mallat, 2012).

The present paper brings these ideas into a single framework centered on operators acting on functions and measures. Dense layers arise when X is a finite set with counting measure. Convolutions arise when X carries translation structure and the kernel depends on differences. Attention arises from normalized state-dependent kernels over an empirical measure. Graph neural networks arise when the kernel is restricted by adjacency, edge weights, or a graph Laplacian. Neural fields arise when the domain is kept continuous. Neural operators arise when the learned map acts between function spaces and is evaluated through discretized input and output fields. Kernel mean methods arise when the object being represented is a probability measure and the readout is a distributional functional.

This formulation separates three objects that are often conflated in finite neural implementations. The first is the operator class, which de-

<https://doi.org/10.1016/j.neunet.2026.109577>

Received 26 April 2026; Received in revised form 2 August 2026; Accepted 29 August 2026

Available online 9 September 2026

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termines the transformations available to the architecture. The second is the represented object, such as a function, measure, graph signal, token measure, or field. The third is the finite computational scheme, such as matrix multiplication, quadrature, projection, sampling, spectral truncation, low-rank approximation, or message passing. This separation makes it possible to compare architectures by the operators they approximate and by the numerical schemes used to approximate them.

The distinction is useful even for ordinary finite-dimensional data. A vector $z \in \mathbb{R}^d$ may be viewed as a scalar field on the coordinate set $\{1, \dots, d\}$. A dense layer is then a kernel operator on a finite measure space. A tabular kernel method is an empirical-measure operator over the training samples. A PCA model is a projection onto an empirical spectral subspace. A Nyström method is a finite-rank approximation of a kernel operator. An image classifier acts on a sampled field over a grid, and convolution, DCT truncation, PCA truncation, and full kernel evaluation give different finite representations of the same underlying image-field problem.

The framework also clarifies the role of approximation. Architecture design becomes the choice of a measure space, function space, kernel class, topology, and discretization scheme suited to the target operator. Numerical analysis supplies quadrature, projection, and stability estimates. Approximation theory supplies representation rates for finite-dimensional approximations. Harmonic analysis supplies bases and transforms adapted to translation, scale, and symmetry. Signal processing supplies efficient finite representations. Measure theory supplies empirical measures, weak convergence, transport distances, kernel embeddings, and distributional functionals.

The present framework is intended to organize the approximation problem rather than to assert a single universal approximation theorem for all of the architectures considered here. Such a theorem necessarily depends on additional choices and assumptions, including the domain and measure, the input and output function spaces, the topology in which approximation is measured, the regularity or compactness of the target class, and the parameterization of the kernel and nonlinear maps. These assumptions differ substantially among finite sets, Euclidean domains, graphs, empirical measures, and function-space operators. Consequently, density results and quantitative approximation rates must be formulated for particular operator classes and representations. The contribution developed here is the common formulation that identifies the objects, error sources, and assumptions from which such architecture-specific results can be derived.

The paper proceeds as follows. Section 2 defines neural layers as operators on functions and measures and shows how finite arrays arise from empirical representations. Section 3 derives dense layers, convolutions, attention mechanisms, graph networks, graphon layers, and neural fields from choices of domain, measure, kernel, and representation. Section 4 studies projection, quadrature, approximation error, stability, topology, and computational structure. Section 5 extends the framework to high-dimensional coordinate fields, product domains, images, point clouds, graphs, token measures, and neural operators. Section 6 gives finite-dimensional experiments on tabular and image data, comparing raw coordinate representations, PCA and DCT truncations, empirical kernel-mean classifiers, Nyström approximations, and exact RBF kernel methods.

2. Neural layers as operators on functions and measures

Let $(X, \mathcal{A})$ be a measurable space and let E be a finite-dimensional normed vector space. Write $\mathcal{M}_+(X)$ for the finite nonnegative measures on X , $\mathcal{P}(X) \subset \mathcal{M}_+(X)$ for the probability measures, and $\mathcal{M}(X; E)$ for the finite E -valued measures. If $\mu \in \mathcal{M}_+(X)$, then $\mathcal{L}_\mu^p(X; E)$ denotes the space of E -valued p -integrable functions with respect to μ .

A neural layer will be treated as an operator whose inputs are either functions $u : X \rightarrow E$, measures $\mu \in \mathcal{M}_+(X)$, or feature-bearing measures

in $\mathcal{M}(X; E)$. Finite arrays appear after choosing a finite-dimensional representation of these objects.

2.1. Finite representations and empirical measures

A finite collection of sites, weights, and features is a triple

$$\{(x_i, a_i, h_i)\}_{i=1}^n, \quad x_i \in X, \quad a_i \geq 0, \quad h_i \in E.$$

It determines a finite measure on X ,

$$\mu_n = \sum_{i=1}^n a_i \delta_{x_i},$$

an E -valued measure on X ,

$$m_n = \sum_{i=1}^n a_i h_i \delta_{x_i},$$

and a lifted measure on $X \times E$,

$$\nu_n = \sum_{i=1}^n a_i \delta_{(x_i, h_i)}.$$

The measure μ_n records the support and weights of the representation. The vector measure m_n records feature mass at each site. The lifted measure ν_n records the joint distribution of sites and features.

For every bounded measurable test function $\varphi : X \rightarrow \mathbb{R}$,

$$\int_X \varphi(x) d\mu_n(x) = \sum_{i=1}^n a_i \varphi(x_i).$$

For every bounded measurable $\psi : X \rightarrow E^*$,

$$\int_X \psi(x) dm_n(x) = \sum_{i=1}^n a_i \psi(x_i)[h_i].$$

Thus sums over tokens, particles, nodes, or sampled grid points are integrals against empirical measures. If $\sum_i a_i = 1$, then $\mu_n \in \mathcal{P}(X)$, and the same formulas describe empirical expectations.

Definition 2.1 (Finite-dimensional representation). A finite-dimensional representation of a feature field or feature measure consists of sites $x_1, \dots, x_n$, weights $a_1, \dots, a_n$, and feature values $h_1, \dots, h_n$. It may be read either as a function u_n on the finite support $\{x_i\}_{i=1}^n$, as the vector measure $m_n = \sum_i a_i h_i \delta_{x_i}$, or as the lifted empirical measure $\nu_n = \sum_i a_i \delta_{(x_i, h_i)}$.

When $X = \{1, \dots, n\}$ and μ is counting measure, a feature field $u : X \rightarrow E$ is exactly a vector

$$(u(1), \dots, u(n)) \in E^n.$$

This is the finite-array case. The same notation also covers nonuniform samples, weighted particles, point clouds, adaptive meshes, and empirical distributions.

2.2. Kernel operators

Let $(Y, \mathcal{B}, \nu)$ be another measure space and let F be a finite-dimensional normed vector space. A measurable kernel with input feature space E and output feature space F is a map

$$K : Y \times X \rightarrow \mathcal{L}(E, F),$$

where $\mathcal{L}(E, F)$ denotes the space of linear maps from E to F . Given $\mu \in \mathcal{M}_+(X)$, the associated kernel operator is

$$(\mathcal{K}_\mu u)(y) = \int_X K(y, x) u(x) d\mu(x).$$

The integral is a Bochner integral. Since E and F are finite-dimensional, this is equivalent to coordinatewise integration.

Proposition 2.2 (Bounded kernel operators). Assume $\mu(X) < \infty$ and

$$\|K\|_{L^\infty} = \operatorname{ess\,sup}_{(y,x) \in Y \times X} \|K(y,x)\|_{\mathcal{L}(E,F)} < \infty.$$

Then $\mathcal{K}_\mu : L^\infty_\mu(X; E) \rightarrow L^\infty_\nu(Y; F)$ is bounded and

$$\|\mathcal{K}_\mu u\|_{L^\infty_\nu} \leq \mu(X) \|K\|_{L^\infty} \|u\|_{L^\infty_\mu}.$$

Proof. For ν -almost every y ,

$$\|\mathcal{K}_\mu u(y)\| \leq \int_X \|K(y,x)\|_{\mathcal{L}(E,F)} \|u(x)\| \, d\mu(x) \leq \mu(X) \|K\|_{L^\infty} \|u\|_{L^\infty_\mu}.$$

Taking the essential supremum over y gives the claim. $\square$

A neural layer is obtained by composing such an aggregation operator with feature maps and pointwise updates. A convenient form is

$$u_{l+1}(y) = \Gamma_l \left(y, u_l(y), \int_X K_l(y, x, u_l) \Psi_l(u_l(x)) \, d\mu_l(x) \right),$$

where $\Psi_l : E_l \rightarrow \tilde{E}_l$ transforms features before aggregation, K_l is an operator-valued kernel, and Γ_l updates the state. The kernel may depend on u_l , which allows attention-type and message-passing-type layers.

If the input is represented empirically by

$$\mu_n = \sum_{i=1}^n a_i \delta_{x_i},$$

then the same layer becomes

$$u_{l+1}(y) = \Gamma_l \left(y, u_l(y), \sum_{i=1}^n a_i K_l(y, x_i, u_l) \Psi_l(u_l(x_i)) \right).$$

Thus finite neural computation is the empirical-measure evaluation of the operator.

Proposition 2.3 (Dense layers as finite-set kernel operators). Let $X = Y = \{1, \dots, n\}$, let μ be counting measure, and let $E = F = \mathbb{R}$. For any matrix $W \in \mathbb{R}^{n \times n}$, define

$$K(i, j) = W_{ij}.$$

Then

$$(\mathcal{K}_\mu h)(i) = \sum_{j=1}^n W_{ij} h(j).$$

Consequently, the dense layer

$$h_{l+1} = \sigma(W h_l + b)$$

is a kernel operator on a finite measure space followed by a pointwise non-linearity.

Proof. Since μ is counting measure,

$$(\mathcal{K}_\mu h)(i) = \int_X K(i, j) h(j) \, d\mu(j) = \sum_{j=1}^n K(i, j) h(j) = \sum_{j=1}^n W_{ij} h(j).$$

Adding $b(i)$ and applying σ pointwise gives the usual dense layer. $\square$

The finite-set case shows that matrices are finite-dimensional representations of kernels. Changing the measure, the domain, or the approximation scheme changes the computation without changing the operator-theoretic form.

2.3. Functionals of measures

Many outputs of neural architectures are scalar or finite-dimensional functionals of a measure. Let $\mu \in \mathcal{M}_+(X)$. A linear readout has the form

$$F_\phi(\mu) = \int_X \phi(x) \, d\mu(x),$$

where $\phi : X \rightarrow \mathbb{R}$ is measurable and integrable with respect to μ . If $\mu = \mu_n$, then

$$F_\phi(\mu_n) = \sum_{i=1}^n a_i \phi(x_i).$$

This includes summation pooling, average pooling, empirical expectations, and moment features.

A quadratic interaction functional has the form

$$E_K(\mu) = \iint_{X \times X} K(x, y) \, d\mu(x) \, d\mu(y).$$

For an empirical measure,

$$E_K(\mu_n) = \sum_{i=1}^n \sum_{j=1}^n a_i a_j K(x_i, x_j).$$

This form includes pairwise energies, kernel losses, graph interaction terms, and distributional discrepancies.

If $k : X \times X \rightarrow \mathbb{R}$ is a positive-definite kernel, the kernel mean embedding of μ is

$$\eta_\mu = \int_X k(x, \cdot) \, d\mu(x) \in \mathcal{H}_k,$$

where $\mathcal{H}_k$ is the reproducing kernel Hilbert space of k . For empirical μ_n ,

$$\eta_{\mu_n} = \sum_{i=1}^n a_i k(x_i, \cdot).$$

Distances between such embeddings give computable discrepancies between measures.

The same formalism also describes maps whose inputs and outputs have different types:

$$\mathcal{T} : \mathcal{F}(X) \rightarrow \mathcal{F}(Y),$$

for operators between feature fields,

$$\mathcal{T} : \mathcal{P}(X) \rightarrow \mathcal{F}(Y),$$

for maps from distributions to functions, and

$$\mathcal{T} : \mathcal{P}(X) \rightarrow \mathcal{P}(Y),$$

for measure-valued transformations. In finite computation these maps are represented by samples, basis coefficients, quadrature rules, or finite arrays.

Proposition 2.4 (Continuity of bounded kernel functionals). Let X be a topological space and let $K \in C_b(X \times X)$. If $\mu_n, \mu \in \mathcal{P}(X)$ and $\mu_n \rightharpoonup \mu$ weakly, then

$$\iint_{X \times X} K(x, y) \, d\mu_n(x) \, d\mu_n(y) \longrightarrow \iint_{X \times X} K(x, y) \, d\mu(x) \, d\mu(y).$$

Proof. Weak convergence $\mu_n \rightharpoonup \mu$ implies product convergence

$$\mu_n \otimes \mu_n \rightharpoonup \mu \otimes \mu$$

on $X \times X$. Since $K \in C_b(X \times X)$, the definition of weak convergence gives

$$\int_{X \times X} K(x, y) \, d(\mu_n \otimes \mu_n)(x, y) \rightarrow \int_{X \times X} K(x, y) \, d(\mu \otimes \mu)(x, y).$$

This is the desired convergence. $\square$

The operator view therefore has two levels. Layers act on feature fields or feature-bearing measures. Readouts, losses, and classifiers are functionals applied to the resulting functions or measures. Finite neural networks are obtained by replacing the integrals and functionals above with their finite-dimensional representations.

3. Standard architectures as special cases

The operator form developed in [Section 2](#) recovers standard neural architectures by specifying the domain, measure, kernel, feature space, and discretization. In each case, the finite computation is obtained by evaluating an integral operator against a finite measure or by projecting a continuous operator onto a finite-dimensional representation.

3.1. Dense layers

Let

$$X = \{1, \dots, n\}, \quad Y = \{1, \dots, m\},$$

and equip both sets with counting measure. Let E and F be finite-dimensional feature spaces. A feature vector

$$h = (h(1), \dots, h(n)) \in E^n$$

is a function $h : X \rightarrow E$. A dense affine layer is specified by linear maps

$$W(i, j) \in \mathcal{L}(E, F), \quad 1 \leq i \leq m, \quad 1 \leq j \leq n,$$

and a bias function $b : Y \rightarrow F$. Define the kernel

$$K(i, j) = W(i, j).$$

The corresponding kernel operator is

$$(\mathcal{K}h)(i) = \int_X K(i, j)h(j) \, d\mu(j) = \sum_{j=1}^n W(i, j)h(j).$$

If $\sigma : F \rightarrow F$ is applied pointwise, the layer is

$$h_{l+1}(i) = \sigma \left(b_l(i) + \sum_{j=1}^n W_l(i, j)h_l(j) \right).$$

Proposition 3.1 (Dense layers are kernel operators on finite measure spaces). *Every dense layer between finite feature arrays is a kernel operator over a finite measure space followed by a pointwise map. Conversely, every kernel operator between finite sets equipped with counting measure is a dense linear layer.*

Proof. Given a dense layer, set $K(i, j) = W(i, j)$. Since the measure on X is counting measure, the integral defining $\mathcal{K}$ is exactly the matrix-vector contraction above. Conversely, any kernel $K : Y \times X \rightarrow \mathcal{L}(E, F)$ over finite sets defines the block matrix with entries $K(i, j)$. Applying the kernel operator is multiplication by this block matrix. $\square$

Thus the dense layer is the case in which the domain has no geometric structure beyond finiteness. Width is the cardinality of the finite representing set, and matrix multiplication is the exact quadrature rule for counting measure.

3.2. Convolutional and harmonic architectures

Let G be a locally compact abelian group with Haar measure dx . The main examples are $G = \mathbb{R}^d$, $G = \mathbb{T}^d$, and finite cyclic grids. Let $u : G \rightarrow E$ be a feature field and let

$$k : G \rightarrow \mathcal{L}(E, F)$$

be an operator-valued convolution kernel. The associated translation-equivariant operator is

$$(\mathcal{K}u)(x) = \int_G k(x - y)u(y) \, dy.$$

Equivalently,

$$\mathcal{K}u = k * u.$$

For a convolutional neural layer one adds a bias and a pointwise nonlinearity:

$$u_{l+1}(x) = \sigma(b_l(x) + (k_l * u_l)(x)).$$

On a finite grid G_n , Haar measure is replaced by normalized or unnormalized counting measure. The convolution becomes

$$(k * u)(x_i) = \sum_j k(x_i - x_j)u(x_j),$$

with the sum restricted according to the chosen stencil, padding rule, or periodic identification. Standard discrete convolutions are therefore finite-dimensional representations of translation-structured kernel operators.

The harmonic representation follows from the Fourier transform. In the scalar case on $\mathbb{R}^d$,

$$\widehat{\mathcal{K}u}(\xi) = \widehat{k}(\xi)\widehat{u}(\xi).$$

For vector-valued features, $\widehat{k}(\xi)$ is a matrix acting on the Fourier coefficient $\widehat{u}(\xi)$:

$$\widehat{\mathcal{K}u}(\xi) = \widehat{k}(\xi)\widehat{u}(\xi), \quad \widehat{k}(\xi) \in \mathcal{L}(E, F).$$

On the torus $\mathbb{T}^d$, the same statement holds for Fourier series:

$$u(x) = \sum_{\xi \in \mathbb{Z}^d} \widehat{u}(\xi)e^{2\pi i \xi \cdot x}, \quad \widehat{\mathcal{K}u}(\xi) = \widehat{k}(\xi)\widehat{u}(\xi).$$

Proposition 3.2 (Translation equivariance of convolution operators). *Let G be a locally compact abelian group and define the translation action*

$$(\tau_a u)(x) = u(x - a).$$

*If $\mathcal{K}u = k * u$, then*

$$\mathcal{K}(\tau_a u) = \tau_a(\mathcal{K}u)$$

for every $a \in G$.

Proof. Using Haar measure invariance,

$$\mathcal{K}(\tau_a u)(x) = \int_G k(x - y)u(y - a) \, dy.$$

Set $z = y - a$. Then

$$\mathcal{K}(\tau_a u)(x) = \int_G k(x - a - z)u(z) \, dz = (\mathcal{K}u)(x - a) = \tau_a(\mathcal{K}u)(x).$$

$\square$

Spectral layers replace the physical-space kernel k by a multiplier $M(\xi)$. A Fourier layer has the form

$$\widehat{u_{l+1}}(\xi) = M_l(\xi)\widehat{u_l}(\xi),$$

possibly followed by inverse transform, bias, and nonlinearity. Fourier neural operators use this idea with a finite set of retained modes:

$$\widehat{u_{l+1}}(\xi) = M_l(\xi)\widehat{u_l}(\xi), \quad \xi \in \Lambda_m \subset \mathbb{Z}^d,$$

while higher modes are discarded, damped, or passed through a separate local term.

Wavelet and scattering constructions use localized multiscale transforms. Given a family of wavelets $\{\psi_\lambda\}_{\lambda \in \Lambda}$, one studies coefficients

$$W_\lambda u(x) = (u * \psi_\lambda)(x).$$

Nonlinear cascades such as

$$u \mapsto |u * \psi_{\lambda_1}| \mapsto ||u * \psi_{\lambda_1}| * \psi_{\lambda_2}|$$

produce stable multiscale features. In the present framework, these are structured operator compositions whose kernels are fixed or partially learned and whose finite-dimensional representation is given by sampled transform coefficients.

3.3. Attention as a data-dependent measure operator

Let a token sequence at layer l be represented by sites and features

$$\{(x_i, h_i)\}_{i=1}^n, \quad h_i \in E_l.$$

The corresponding feature-bearing empirical measure is

$$\nu_l = \sum_{i=1}^n \delta_{(x_i, h_i)}.$$

Let

$$Q_l : E_l \rightarrow \mathbb{R}^r, \quad K_l : E_l \rightarrow \mathbb{R}^r, \quad V_l : E_l \rightarrow F_l$$

be query, key, and value maps. For a query site x_i , define a probability measure on the token set by

$$\pi_l(\{x_j\} \mid x_i) = \frac{\exp(Q_l h_i \cdot K_l h_j)}{\sum_{m=1}^n \exp(Q_l h_i \cdot K_l h_m)}.$$

Then self-attention is the conditional expectation

$$h_{l+1}(x_i) = \int_X V_l h_l(y) \, d\pi_l(y | x_i).$$

Since $\pi_l(\cdot | x_i)$ is supported on the empirical token set, this becomes

$$h_{l+1}(x_i) = \sum_{j=1}^n \alpha_{ij}^{(l)} V_l h_j,$$

where

$$\alpha_{ij}^{(l)} = \frac{\exp(Q_l h_i \cdot K_l h_j)}{\sum_{m=1}^n \exp(Q_l h_i \cdot K_l h_m)}.$$

This is a data-dependent Markov kernel. For each x_i ,

$$\alpha_{ij}^{(l)} \geq 0, \quad \sum_{j=1}^n \alpha_{ij}^{(l)} = 1.$$

Thus attention sends each token to an average of value features with respect to a learned probability measure depending on the current state.

A continuum analogue is obtained by replacing the empirical measure with $\mu_l \in \mathcal{P}(X)$. Suppose

$$q_l : X \rightarrow \mathbb{R}^r, \quad k_l : X \rightarrow \mathbb{R}^r, \quad v_l : X \rightarrow F_l$$

are measurable functions determined by u_l . Define

$$Z_l(x) = \int_X \exp(q_l(x) \cdot k_l(y)) \, d\mu_l(y),$$

assuming $0 < Z_l(x) < \infty$. Then

$$d\pi_l(y | x) = \frac{\exp(q_l(x) \cdot k_l(y))}{Z_l(x)} \, d\mu_l(y),$$

and the attention operator is

$$(\mathcal{A}_l u_l)(x) = \int_X v_l(y) \, d\pi_l(y | x).$$

The empirical transformer layer is the quadrature rule associated with the empirical measure $\mu_l = \sum_j \delta_{x_j}$.

Proposition 3.3 (Attention is a normalized kernel operator). *For fixed u_l , the attention operator is a kernel operator with kernel*

$$A_l(x, y; u_l) = \frac{\exp(q_l(x) \cdot k_l(y))}{Z_l(x)}.$$

It is normalized in the sense that

$$\int_X A_l(x, y; u_l) \, d\mu_l(y) = 1$$

for every x for which $Z_l(x)$ is finite and nonzero.

Proof. The expression for $\mathcal{A}_l$ can be written as

$$(\mathcal{A}_l u_l)(x) = \int_X A_l(x, y; u_l) v_l(y) \, d\mu_l(y).$$

The normalization follows directly from the definition of $Z_l(x)$. $\square$

The quadratic cost of full finite attention comes from evaluating all pairwise kernel values $A_l(x_i, x_j; u_l)$. Low-rank attention, sparse attention, random-feature attention, locality restrictions, and Nyström approximations are different finite-dimensional representations or approximations of this normalized kernel operator.

3.4. Graphs, graphons, and neural fields

Let $G_n = (V_n, E_n)$ be a finite graph with $V_n = \{1, \dots, n\}$. A graph signal is a function

$$h : V_n \rightarrow E.$$

Let $A \in \mathbb{R}^{n \times n}$ be a weighted adjacency matrix and let D be the degree matrix. A basic graph message-passing layer has the form

$$h_{l+1}(i) = \sigma\left(b_l(i) + \sum_{j=1}^n A_{ij} W_l h_l(j)\right).$$

This is a finite-set kernel operator with

$$K_l(i, j) = A_{ij} W_l.$$

Normalized graph convolution uses kernels such as

$$K_l(i, j) = (D^{-1/2} A D^{-1/2})_{ij} W_l$$

or corresponding Laplacian-based spectral multipliers.

Graph-spectral architectures use the eigendecomposition of a graph Laplacian L . If

$$L \varphi_k = \lambda_k \varphi_k,$$

then a spectral graph filter has the form

$$g(L)h = \sum_k g(\lambda_k) \langle h, \varphi_k \rangle \varphi_k.$$

Polynomial filters,

$$g(L) = \sum_{r=0}^R c_r L^r,$$

give localized computations because powers of L propagate information through bounded graph neighborhoods.

Graphon models replace finite adjacency matrices by measurable kernels. Let

$$W : [0, 1]^2 \rightarrow [0, 1]$$

be a symmetric measurable function. The associated graphon operator is

$$(\mathcal{W}f)(x) = \int_0^1 W(x, y) f(y) \, dy.$$

A graphon neural layer has the form

$$u_{l+1}(x) = \sigma\left(b_l(x) + \int_0^1 W_l(x, y) A_l u_l(y) \, dy\right),$$

where A_l is a feature map or linear feature transform. Finite graph neural networks are obtained by sampling points $x_1, \dots, x_n \in [0, 1]$, forming an empirical approximation to $W(x_i, x_j)$, and evaluating the graphon integral by a finite sum.

Neural fields arise when the unit index is kept continuous. Let X be a spatial, sensory, or feature domain equipped with a measure μ . A standard neural-field equation is

$$\tau \partial_t u(x, t) = -u(x, t) + \sigma\left(\int_X K(x, y) u(y, t) \, d\mu(y) + I(x, t)\right).$$

The term $-u(x, t)$ gives decay, $I(x, t)$ is an external input, and the integral term gives nonlocal interaction across the domain. Spatial discretization at sites $x_1, \dots, x_n$ with weights $a_1, \dots, a_n$ gives the finite system

$$\tau \frac{d}{dt} u_i(t) = -u_i(t) + \sigma\left(\sum_{j=1}^n a_j K(x_i, x_j) u_j(t) + I(x_i, t)\right),$$

where $u_i(t) = u(x_i, t)$. Thus recurrent finite neural systems can be obtained as quadrature discretizations of neural-field equations.

Proposition 3.4 (Discretization of neural fields). *Let $x_1, \dots, x_n \in X$ and $a_1, \dots, a_n \geq 0$ define an empirical measure*

$$\mu_n = \sum_{j=1}^n a_j \delta_{x_j}.$$

Replacing μ by μ_n in the neural-field equation yields the finite-dimensional dynamical system

$$\tau \frac{d}{dt} u_i(t) = -u_i(t) + \sigma\left(\sum_{j=1}^n a_j K(x_i, x_j) u_j(t) + I(x_i, t)\right).$$

Proof. Evaluate the field equation at $x = x_i$ and replace the integral against μ by the integral against μ_n :

$$\int_X K(x_i, y) u(y, t) \, d\mu_n(y) = \sum_{j=1}^n a_j K(x_i, x_j) u(x_j, t).$$

Setting $u_j(t) = u(x_j, t)$ gives the stated system. $\square$

Dense networks, convolutions, attention mechanisms, graph networks, graphon operators, and neural fields are therefore obtained by changing the measure space, kernel structure, symmetry constraints, and finite-dimensional representation used to evaluate the same basic operator form.

4. Approximation, stability, and computation

The operator formulation becomes a computation only after choosing a finite-dimensional representation. This choice has three parts: a finite space in which states are represented, a rule for approximating integrals, and a finite parameterization of the operator. The resulting architecture is a computable approximation of an operator acting on functions or measures.

4.1. Projection and discretization

Let $\mathcal{F}(X; E)$ and $\mathcal{G}(Y; F)$ be normed spaces of feature fields. Let

$$\mathcal{T} : \mathcal{F}(X; E) \rightarrow \mathcal{G}(Y; F)$$

be the target operator. A finite-dimensional representation of the input space is given by a subspace

$$V_m = \text{span}\{\phi_1, \dots, \phi_m\} \subset \mathcal{F}(X; E)$$

and a projection or approximation map

$$P_m : \mathcal{F}(X; E) \rightarrow V_m.$$

For $u \in \mathcal{F}(X; E)$, the represented field is

$$u_m = P_m u = \sum_{k=1}^m c_k(u) \phi_k.$$

The coefficients $c_1(u), \dots, c_m(u)$ are the finite-dimensional representation of u . Depending on the chosen representation, these coefficients may be nodal values, Fourier coefficients, wavelet coefficients, finite-element coefficients, particle weights, or learned latent coordinates.

If the operator contains an integral,

$$(\mathcal{K}u)(y) = \int_X K(y, x)u(x) \, d\mu(x),$$

then a finite computation also requires a quadrature or sampling rule. A quadrature rule is a finite signed or nonnegative measure

$$\mu_n = \sum_{i=1}^n w_i \delta_{x_i}$$

used in place of μ . The discretized operator is

$$(\mathcal{K}_n u)(y) = \int_X K(y, x)u(x) \, d\mu_n(x) = \sum_{i=1}^n w_i K(y, x_i)u(x_i).$$

If the output is also represented in a finite-dimensional space $W_r \subset \mathcal{G}(Y; F)$ with projection $R_r : \mathcal{G}(Y; F) \rightarrow W_r$, then the fully represented operator is

$$\mathcal{T}_{r,n,m} = R_r \mathcal{T}_n P_m.$$

Thus the computable layer is obtained by input projection, finite evaluation of the operator, and output projection.

For a nonlinear kernel layer of the form

$$(\mathcal{T}u)(y) = \Gamma\left(y, u(y), \int_X K(y, x, u)\Psi(u(x)) \, d\mu(x)\right),$$

the corresponding finite evaluation is

$$(\mathcal{T}_n u_m)(y) = \Gamma\left(y, u_m(y), \sum_{i=1}^n w_i K(y, x_i, u_m)\Psi(u_m(x_i))\right).$$

This formula includes grid discretizations, collocation methods, particle methods, attention over empirical measures, graph message passing, and finite-element quadrature.

Definition 4.1 (Consistent finite representation). Let $\mathcal{F}(X; E)$ be a normed space. A sequence of projections $P_m : \mathcal{F}(X; E) \rightarrow V_m$ is consistent on a subset $\mathcal{U} \subset \mathcal{F}(X; E)$ if

$$\lim_{m \rightarrow \infty} \|u - P_m u\|_{\mathcal{F}} = 0$$

for every $u \in \mathcal{U}$.

Definition 4.2 (Consistent quadrature). Let $\mathcal{C}$ be a class of integrable scalar test functions on X . A sequence of finite measures

$$\mu_n = \sum_{i=1}^n w_i \delta_{x_i}$$

is consistent for μ on $\mathcal{C}$ if

$$\lim_{n \rightarrow \infty} \left| \int_X f \, d\mu - \sum_{i=1}^n w_i f(x_i) \right| = 0$$

for every $f \in \mathcal{C}$.

The relevant test class $\mathcal{C}$ is determined by the operator. For a kernel operator, $\mathcal{C}$ contains functions of the form

$$x \mapsto K(y, x)u(x)$$

or, in the nonlinear case,

$$x \mapsto K(y, x, u)\Psi(u(x)).$$

Quadrature accuracy must therefore be measured on the functions actually produced by the layer.

4.2. Operator error and the sources of approximation rates

Let $\mathcal{T} : \mathcal{F}(X; E) \rightarrow \mathcal{G}(Y; F)$ be the target operator and let $\mathcal{T}_{\theta,n,m}$ be a computable approximation depending on parameters θ , input representation dimension m , and quadrature size n . For $u \in \mathcal{F}(X; E)$,

$$\|\mathcal{T}u - \mathcal{T}_{\theta,n,m} P_m u\|_{\mathcal{G}}$$

is the operator error at u . A useful decomposition is obtained by inserting intermediate approximations:

$$\begin{aligned} \|\mathcal{T}u - \mathcal{T}_{\theta,n,m} P_m u\|_{\mathcal{G}} &\leq \|\mathcal{T}u - \mathcal{T} P_m u\|_{\mathcal{G}} \\ &\quad + \|\mathcal{T} P_m u - \mathcal{T}_{\theta} P_m u\|_{\mathcal{G}} \\ &\quad + \|\mathcal{T}_{\theta} P_m u - \mathcal{T}_{\theta,n} P_m u\|_{\mathcal{G}} \\ &\quad + \|\mathcal{T}_{\theta,n} P_m u - \mathcal{T}_{\theta,n,m} P_m u\|_{\mathcal{G}}. \end{aligned}$$

The four terms correspond to representation error, operator-class error, quadrature or sampling error, and output representation error. If θ is obtained by training, there is also optimization error, measured at the level of the loss.

This decomposition is an organizational identity rather than, by itself, a universal convergence or density result. To obtain a theorem in which the right-hand side vanishes, one must separately establish that the chosen projections converge in the relevant input and output spaces, that the parameterized operator class is dense in an appropriate target class, that the quadrature or sampling rule is consistent for the induced integrands, and that the optimization procedure controls the represented objective. Quantitative rates additionally depend on properties such as smoothness, kernel regularity, spectral decay, geometric dimension, sampling design, and the topology used to measure error. The framework makes these assumptions explicit and separates their contributions, while architecture-specific density theorems and sharp rates require further specialization.

For a data distribution ρ on $\mathcal{F}(X; E)$ and a loss functional $\mathcal{L}$, the population objective is

$$\mathcal{R}(\theta) = \int \mathcal{L}(\mathcal{T}_{\theta,n,m} P_m u, \mathcal{T}u) \, d\rho(u).$$

If $\hat{\theta}$ is the parameter returned by an optimization procedure and θ^* minimizes the represented objective over the chosen parameter class, then

$$E_{\text{opt}} = \mathcal{R}(\hat{\theta}) - \mathcal{R}(\theta^*)$$

measures optimization error. The full error depends on both analytic approximation and the ability of training to find a good parameter.

Proposition 4.3 (Representation error under Lipschitz continuity). *Let $\mathcal{T} : F(X; E) \rightarrow \mathcal{G}(Y; F)$ be Lipschitz on a set $\mathcal{U} \subset F(X; E)$, so that*

$$\|\mathcal{T}u - \mathcal{T}v\|_{\mathcal{G}} \leq L\|u - v\|_F$$

for all $u, v \in \mathcal{U}$. If $P_m u \in \mathcal{U}$, then

$$\|\mathcal{T}u - \mathcal{T}P_m u\|_{\mathcal{G}} \leq L\|u - P_m u\|_F.$$

Proof. The claim follows directly from the Lipschitz condition with $v = P_m u$. $\square$

For kernel operators, quadrature error can be stated directly. Let

$$(\mathcal{K}u)(y) = \int_X K(y, x)u(x) \, d\mu(x), \quad (\mathcal{K}_n u)(y) = \sum_{i=1}^n w_i K(y, x_i)u(x_i).$$

Then

$$\|\mathcal{K}u - \mathcal{K}_n u\|_{L^\infty(Y; F)} \leq \sup_{y \in Y} \left\| \int_X K(y, x)u(x) \, d\mu(x) - \sum_{i=1}^n w_i K(y, x_i)u(x_i) \right\|.$$

The right-hand side is exactly the quadrature error for the family of functions

$$\{x \mapsto K(y, x)u(x) : y \in Y\}.$$

Approximation rates depend on the representation. Smooth periodic functions may have rapidly decaying Fourier coefficients. Functions with localized singularities may be better represented by wavelets or adaptive finite elements. Operators with smooth or low-rank kernels may admit separated approximations. Empirical measures may be approximated by Monte Carlo sampling, deterministic quadrature, sparse grids, or transport-based quantization. The correct rate is therefore tied to the regularity of the input, the structure of the kernel, and the topology used to measure error.

4.3. Stability and topology

A finite-dimensional representation is useful only if small perturbations in the represented object do not produce uncontrolled changes in the output. Stability requires specifying a topology on the input and output spaces.

For measures, common choices include weak convergence, Wasserstein distance, total variation, and kernel-induced metrics. Weak convergence is defined by

$$\mu_n \rightarrow \mu$$

when

$$\int_X f \, d\mu_n \rightarrow \int_X f \, d\mu$$

for every bounded continuous $f : X \rightarrow \mathbb{R}$. Wasserstein distances incorporate the geometry of X . Total variation gives a strong norm on measures. Kernel metrics such as maximum mean discrepancy compare measures through a class of test functions determined by a reproducing kernel.

For fields, the topology may come from L^p , Sobolev, Besov, bounded variation, or reproducing-kernel Hilbert norms. These choices encode different forms of regularity. Sobolev norms control weak derivatives. Besov norms capture multiscale smoothness. RKHS norms control complexity relative to a positive-definite kernel.

Proposition 4.4 (Stability of bounded linear kernel operators). *Let $\mu \in \mathcal{M}_+(X)$, let $K \in L^\infty(Y \times X; \mathcal{L}(E, F))$, and define*

$$(\mathcal{K}u)(y) = \int_X K(y, x)u(x) \, d\mu(x).$$

Then

$$\|\mathcal{K}u - \mathcal{K}v\|_{L^\infty(Y; F)} \leq \mu(X)\|K\|_{L^\infty}\|u - v\|_{L^\infty(X; E)}.$$

Proof. For every y ,

$$\begin{aligned} \|(\mathcal{K}u)(y) - (\mathcal{K}v)(y)\| &= \left\| \int_X K(y, x)(u(x) - v(x)) \, d\mu(x) \right\| \\ &\leq \int_X \|K(y, x)\|_{\mathcal{L}(E, F)} \|u(x) - v(x)\| \, d\mu(x) \\ &\leq \mu(X)\|K\|_{L^\infty}\|u - v\|_{L^\infty(X; E)}. \end{aligned}$$

Taking the supremum over y gives the result. $\square$

For measure inputs, stability is often expressed through continuity of functionals. If $K \in C_b(X \times X)$, then

$$\mu \mapsto \iint_{X \times X} K(x, y) \, d\mu(x) \, d\mu(y)$$

is continuous under weak convergence of probability measures. If K is Lipschitz and X is a metric space, stronger bounds can be stated in Wasserstein distance.

Proposition 4.5 (Wasserstein stability of linear readouts). *Let (X, d) be a metric space and let $\phi : X \rightarrow \mathbb{R}$ be L_ϕ -Lipschitz. If $\mu, \nu \in \mathcal{P}(X)$ have finite first moments, then*

$$\left| \int_X \phi \, d\mu - \int_X \phi \, d\nu \right| \leq L_\phi W_1(\mu, \nu).$$

Proof. This is the Kantorovich–Rubinstein dual bound for W_1 . Since ϕ/L_ϕ is 1-Lipschitz when $L_\phi > 0$, the stated inequality follows. The case $L_\phi = 0$ is immediate. $\square$

The topology must match the operator. Weak convergence is appropriate for bounded continuous test functions. Wasserstein metrics are appropriate when geometry matters. Total variation is appropriate when pointwise mass changes must be controlled. Sobolev and Besov norms are appropriate when derivatives, weak forms, or multiscale regularity matter. RKHS metrics are appropriate when measures are compared through kernel evaluations. A stable architecture is one whose operator is continuous in the topology induced by the representation and by the task.

4.4. Computational structure

Different finite-dimensional representations expose different computational structure. The same operator may be expensive in one representation and cheap in another.

- **Fourier representations.** Translation-invariant operators on $\mathbb{R}^d$, $\mathbb{T}^d$, or finite periodic grids become multiplication operators in frequency space. If

$$\mathcal{K}u = k * u,$$

then

$$\widehat{\mathcal{K}u}(\xi) = \widehat{k}(\xi)\widehat{u}(\xi).$$

Fast Fourier transforms give efficient application on regular grids. The representation is strongest when boundary effects, nonuniform geometry, and localized singularities are controlled.

- **Wavelet representations.** Wavelet bases and frames represent functions through scale and location. Operators with local or multiscale structure often have sparse or compressible matrices in a wavelet representation. This is useful for signals with localized features, discontinuities, or scale-dependent behavior.

- **Finite-element representations.** Finite-element spaces use local basis functions adapted to a mesh. They are natural for bounded domains, irregular geometry, boundary conditions, and weak formulations of differential or integral equations. Locality of the basis often produces sparse matrices.
- **Low-rank kernel representations.** If an integral kernel admits an approximation

$$K(x, y) \approx \sum_{r=1}^R a_r(x) b_r(y),$$

then

$$\int_X K(x, y) u(y) d\mu(y) \approx \sum_{r=1}^R a_r(x) \int_X b_r(y) u(y) d\mu(y).$$

Pairwise interaction is replaced by a finite number of feature integrals. This is the mechanism behind separated expansions, Nyström approximations, random features, and many efficient attention approximations.

- **Particle and quadrature representations.** Empirical measures approximate integrals by weighted samples:

$$\int_X f(x) d\mu(x) \approx \sum_{i=1}^n w_i f(x_i).$$

The error is controlled by the sampling rule and the regularity of the test class. Monte Carlo methods give probabilistic rates. Deterministic quadrature can give faster rates when the dimension and regularity permit it.

- **Graph-spectral representations.** Graph Laplacian eigenvectors provide a frequency representation for graph signals. A spectral graph filter has the form

$$g(L)h = \sum_k g(\lambda_k) \langle h, \varphi_k \rangle \varphi_k.$$

Polynomial filters avoid full eigendecomposition and produce localized computations through powers of L .

- **Kernel-embedding representations.** A positive-definite kernel k embeds a measure into an RKHS by

$$\eta_\mu = \int_X k(x, \cdot) d\mu(x).$$

For an empirical measure,

$$\eta_{\mu_n} = \sum_{i=1}^n w_i k(x_i, \cdot).$$

Inner products between embeddings reduce distributional comparison to kernel evaluations.

- **Adaptive representations.** Adaptive meshes, learned dictionaries, sparse bases, and particle resampling change the finite-dimensional representation during computation. These methods are useful when the relevant degrees of freedom are concentrated in a small region of the domain, frequency spectrum, or feature space.

The choice of representation determines the cost of applying the operator. A dense kernel on n empirical sites has pairwise cost of order n^2 . A local kernel has cost proportional to the number of nonzero interactions. A convolutional kernel on a regular grid can be applied by FFT methods. A low-rank kernel with rank R reduces pairwise interaction to R feature integrals. A sparse finite-element operator has cost proportional to the number of mesh neighbors.

The computational problem is therefore to choose a representation in which the target operator is stable, accurate, and cheap to evaluate. This is the point at which architecture design becomes numerical approximation: the architecture specifies an operator class, and computation specifies the finite-dimensional representation through which that operator is evaluated.

5. High-dimensional extensions

The operator formulation extends directly to high-dimensional data by changing the underlying domain, measure, kernel class, and approximation scheme. A data point is no longer treated as a vector in isolation. It is treated as a finite representation of a field, a measure, or an operator input over a structured high-dimensional domain.

5.1. Coordinate fields in high dimension

Let a tabular observation be

$$z = (z_1, \dots, z_d) \in \mathbb{R}^d.$$

Instead of identifying z only with a point in Euclidean space, define a coordinate domain

$$X_d = \{1, \dots, d\}$$

with counting measure, and represent z as a scalar field

$$u_z : X_d \rightarrow \mathbb{R}, \quad u_z(j) = z_j.$$

A linear layer is the finite kernel operator

$$(Ku_z)(i) = \sum_{j=1}^d K(i, j) u_z(j), \quad K(i, j) \in \mathbb{R}.$$

If K is arbitrary, this is a dense layer. If K has imposed structure, such as sparsity, low rank, bandedness, block structure, or hierarchical structure, the architecture encodes prior information about feature interactions.

For vector-valued coordinate features,

$$u_z : X_d \rightarrow E,$$

the kernel becomes

$$K(i, j) \in \mathcal{L}(E, F),$$

and the layer is

$$(Ku_z)(i) = \sum_{j=1}^d K(i, j) u_z(j).$$

This covers learned feature embeddings for categorical variables, grouped covariates, multi-channel sensor readings, and mixed tabular data.

A nonlinear coordinate operator has the form

$$u_{l+1}(i) = \Gamma_l \left(i, u_l(i), \sum_{j=1}^d K_l(i, j, u_l) \Psi_l(u_l(j)) \right).$$

Feature-crossing layers, gated tabular models, coordinate attention, and dense multilayer networks are all finite-dimensional examples of this form.

5.2. High-dimensional product domains

Many high-dimensional inputs are fields over product spaces. Let

$$X = X_1 \times \dots \times X_s, \quad \mu = \mu_1 \otimes \dots \otimes \mu_s.$$

An image has $X = \{1, \dots, H\} \times \{1, \dots, W\}$. A video has

$$X = \{1, \dots, T\} \times \{1, \dots, H\} \times \{1, \dots, W\}.$$

A volumetric scan has

$$X = \{1, \dots, H\} \times \{1, \dots, W\} \times \{1, \dots, D\}.$$

A spatiotemporal physical field has

$$X = [0, T] \times D, \quad D \subset \mathbb{R}^r.$$

A feature field is

$$u : X \rightarrow E.$$

A general kernel operator is

$$(\mathcal{K}u)(x) = \int_X K(x, y)u(y) \, d\mu(y), \quad K(x, y) \in \mathcal{L}(E, F).$$

For a finite grid

$$X_h = X_{1,h} \times \cdots \times X_{s,h},$$

with quadrature weights w_y , the computable operator is

$$(\mathcal{K}_h u)(x) = \sum_{y \in X_h} w_y K(x, y)u(y).$$

The direct dense kernel over $N = \prod_{r=1}^s |X_{r,h}|$ sites has cost of order N^2 . Structured kernels reduce this cost.

A separable kernel has the form

$$K(x, y) = \sum_{q=1}^R A_q \prod_{r=1}^s k_{q,r}(x_r, y_r), \quad A_q \in \mathcal{L}(E, F).$$

Then

$$(\mathcal{K}u)(x) = \sum_{q=1}^R A_q \int_X \left(\prod_{r=1}^s k_{q,r}(x_r, y_r) \right) u(y) \, d\mu(y).$$

This representation underlies tensor-product filters, separable convolutions, low-rank integral operators, spectral tensor methods, and efficient neural operators.

Proposition 5.1 (Error from separable kernel approximation). *Let X have finite measure and let*

$$K, K_R : X \times X \rightarrow \mathcal{L}(E, F)$$

be bounded measurable kernels. Define

$$(\mathcal{K}u)(x) = \int_X K(x, y)u(y) \, d\mu(y), \quad (\mathcal{K}_R u)(x) = \int_X K_R(x, y)u(y) \, d\mu(y).$$

If

$$\Delta_R = \operatorname{ess\,sup}_{x,y \in X} \|K(x, y) - K_R(x, y)\|_{\mathcal{L}(E, F)},$$

then

$$\|\mathcal{K}u - \mathcal{K}_R u\|_{L^\infty(X; F)} \leq \mu(X) \Delta_R \|u\|_{L^\infty(X; E)}.$$

Proof. For almost every $x \in X$,

$$\begin{aligned} \|(\mathcal{K} - \mathcal{K}_R)u(x)\| &= \left\| \int_X (K(x, y) - K_R(x, y))u(y) \, d\mu(y) \right\| \\ &\leq \int_X \|K(x, y) - K_R(x, y)\|_{\mathcal{L}(E, F)} \|u(y)\| \, d\mu(y) \\ &\leq \mu(X) \Delta_R \|u\|_{L^\infty(X; E)}. \end{aligned}$$

Taking the essential supremum over x gives the claim. $\square$

This proposition gives a uniform bound for tensor-train kernels, canonical polyadic kernels, Nyström kernels, random-feature kernels, and low-rank attention kernels.

5.3. Images, volumes, and spatiotemporal fields

Let

$$\Omega_h = \{1, \dots, H\} \times \{1, \dots, W\}$$

be an image grid. A multi-channel image is a field

$$u : \Omega_h \rightarrow \mathbb{R}^c.$$

A dense image operator has the form

$$(\mathcal{K}u)(p) = \sum_{q \in \Omega_h} K(p, q)u(q).$$

A convolutional operator imposes translation structure:

$$K(p, q) = k(p - q),$$

so that

$$(\mathcal{K}u)(p) = \sum_{q \in \Omega_h} k(p - q)u(q).$$

A local convolution imposes compact support:

$$k(r) = 0 \quad \text{for } \|r\|_\infty > R.$$

The cost then scales with

$$|\Omega_h|(2R + 1)^2$$

instead of

$$|\Omega_h|^2.$$

For a video,

$$u : \{1, \dots, T\} \times \Omega_h \rightarrow \mathbb{R}^c,$$

a spatiotemporal kernel has the form

$$(\mathcal{K}u)(t, p) = \sum_{s=1}^T \sum_{q \in \Omega_h} K(t, p, s, q)u(s, q).$$

A separable spatiotemporal kernel can be written as

$$K(t, p, s, q) = \sum_{r=1}^R a_r(t, s) b_r(p, q),$$

which gives

$$(\mathcal{K}u)(t, p) = \sum_{r=1}^R \sum_{s=1}^T a_r(t, s) \sum_{q \in \Omega_h} b_r(p, q)u(s, q).$$

This expresses temporal filtering and spatial filtering as coupled low-rank operator factors.

For volumetric data,

$$u : \{1, \dots, H\} \times \{1, \dots, W\} \times \{1, \dots, D\} \rightarrow E.$$

The same construction applies with kernels over three spatial coordinates. Local kernels, separable kernels, Fourier multipliers, wavelet transforms, and multigrid representations give different finite approximations of the same operator class.

5.4. High-dimensional spectral representations

Let $X = \mathbb{T}^s$, and let

$$u : X \rightarrow E.$$

The Fourier expansion is

$$u(x) = \sum_{\xi \in \mathbb{Z}^s} \hat{u}(\xi) e^{2\pi i \xi \cdot x}.$$

A translation-invariant operator satisfies

$$\widehat{\mathcal{K}u}(\xi) = M(\xi) \hat{u}(\xi), \quad M(\xi) \in \mathcal{L}(E, F).$$

A finite spectral representation chooses a frequency set

$$\Lambda_m \subset \mathbb{Z}^s$$

and uses

$$P_m u = \sum_{\xi \in \Lambda_m} \hat{u}(\xi) e^{2\pi i \xi \cdot x}.$$

The represented operator is

$$\mathcal{K}_m u = \sum_{\xi \in \Lambda_m} M(\xi) \hat{u}(\xi) e^{2\pi i \xi \cdot x}.$$

If u has Sobolev regularity, spectral truncation gives quantitative approximation rates. Let

$$\|u\|_{H^s(\mathbb{T}^s)}^2 = \sum_{\xi \in \mathbb{Z}^s} (1 + \|\xi\|^2)^s \|\hat{u}(\xi)\|^2.$$

For

$$\Lambda_m = \{\xi \in \mathbb{Z}^r : \|\xi\| \leq m\},$$

one has

$$\|u - P_m u\|_{L^2(\mathbb{T}^r)} \leq m^{-s} \|u\|_{H^s(\mathbb{T}^r)}.$$

Thus spectral neural operators are most efficient when the target fields have decay in frequency. High-dimensional smoothness, anisotropy, and sparsity determine whether the number of retained modes is feasible.

Anisotropic truncation uses sets of the form

$$\Lambda_m = \{\xi \in \mathbb{Z}^r : |\xi_j| \leq m_j, 1 \leq j \leq r\}.$$

This is appropriate when regularity differs by coordinate, as in spatiotemporal data where time and space have different scales. Hyperbolic-cross truncation uses

$$\Lambda_m^{\text{hc}} = \left\{ \xi \in \mathbb{Z}^r : \prod_{j=1}^r (1 + |\xi_j|) \leq m \right\},$$

which is useful for mixed smoothness classes and reduces the number of retained modes in high dimension.

5.5. Point clouds and empirical measures in high dimension

Let

$$x_1, \dots, x_n \in \mathbb{R}^d$$

be a point cloud. The natural representation is the empirical measure

$$\mu_n = \frac{1}{n} \sum_{i=1}^n \delta_{x_i}.$$

With features $h_i \in E$, define

$$\nu_n = \frac{1}{n} \sum_{i=1}^n \delta_{(x_i, h_i)}.$$

A permutation-invariant map has the form

$$F(\nu_n) = \rho \left(\frac{1}{n} \sum_{i=1}^n \phi(x_i, h_i) \right).$$

A permutation-equivariant kernel layer has the form

$$h'_i = \Gamma \left(x_i, h_i, \frac{1}{n} \sum_{j=1}^n K(x_i, x_j, h_i, h_j) \Psi(h_j) \right).$$

The domain dimension d influences both geometry and sample complexity. If the points lie near a lower-dimensional set $M \subset \mathbb{R}^d$, kernels should be adapted to the intrinsic geometry of M . A local geometric kernel can be written as

$$K_\epsilon(x, y) = \eta \left(\frac{\|x - y\|^2}{\epsilon} \right) A(x, y),$$

where η is a decaying profile and $A(x, y) \in \mathcal{L}(E, F)$. The bandwidth ϵ determines the interaction scale. As n grows and ϵ shrinks, graph Laplacian and diffusion-type operators can approximate continuum differential operators on the underlying domain.

A distributional readout can use kernel mean embeddings:

$$\eta_{\mu_n} = \frac{1}{n} \sum_{i=1}^n k(x_i, \cdot).$$

If k is characteristic, the embedding separates probability measures. A classifier acting on point clouds can be written as

$$f(\mu_n) = g(\eta_{\mu_n}).$$

For finite computation, one may approximate k by random features:

$$k(x, y) \approx \frac{1}{R} \sum_{r=1}^R \varphi_r(x) \varphi_r(y),$$

giving

$$\eta_{\mu_n} \approx \frac{1}{n} \sum_{i=1}^n \frac{1}{\sqrt{R}} (\varphi_1(x_i), \dots, \varphi_R(x_i)).$$

The representation dimension is R , independent of n .

5.6. Graphs with high-dimensional node and edge features

Let $G = (V, E_G)$ be a graph with $|V| = n$. A node field is

$$h : V \rightarrow E.$$

If edges have features $e_{ij} \in B$, a message-passing layer is

$$h'_i = \Gamma \left(h_i, \sum_{j \in \mathcal{N}(i)} K(h_i, h_j, e_{ij}) \Psi(h_j) \right).$$

This is a kernel operator over the finite measure

$$\mu_i = \sum_{j \in \mathcal{N}(i)} a_{ij} \delta_j,$$

where a_{ij} may be fixed, normalized, learned, or attention-based.

For graph-level prediction, the graph is represented by a feature-bearing empirical measure

$$\nu_G = \frac{1}{n} \sum_{i=1}^n \delta_{(i, h_i)}$$

together with an interaction kernel determined by edges. Pooling is integration:

$$\bar{h} = \frac{1}{n} \sum_{i=1}^n h_i = \int h \, d\nu_G.$$

Spectral graph methods use the Laplacian

$$L = I - D^{-1/2} A D^{-1/2}.$$

If

$$L \varphi_k = \lambda_k \varphi_k,$$

a spectral filter is

$$g(L)h = \sum_{k=1}^n g(\lambda_k) \langle h, \varphi_k \rangle \varphi_k.$$

Polynomial filters

$$g(L) = \sum_{r=0}^R c_r L^r$$

give local operators with R -hop support. This is the graph analogue of approximating a continuum operator by a finite spectral or polynomial expansion.

For graph sequences converging to a graphon W , the finite adjacency operators approximate

$$(\mathcal{W}f)(x) = \int_0^1 W(x, y) f(y) \, dy.$$

This gives a route to resolution-independent graph learning.

5.7. Attention on high-dimensional token measures

A token sequence with positions $i \in \{1, \dots, n\}$ and embeddings $h_i \in E$ can be represented as

$$\nu_n = \frac{1}{n} \sum_{i=1}^n \delta_{(p_i, h_i)},$$

where p_i is a positional feature. Self-attention defines a state-dependent Markov kernel

$$\pi(j | i) = \frac{\exp(q(h_i, p_i) \cdot k(h_j, p_j))}{\sum_{m=1}^n \exp(q(h_i, p_i) \cdot k(h_m, p_m))}.$$

The update is

$$h'_i = \sum_{j=1}^n \pi(j | i) v(h_j, p_j).$$

In high dimension, the cost of full attention is order n^2 . Low-rank attention approximates the kernel

$$A(i, j) = \exp(q_i \cdot k_j)$$

by

$$A(i, j) \approx \sum_{r=1}^R a_r(i) b_r(j).$$

Then

$$h'_i \approx \sum_{r=1}^R a_r(i) \sum_{j=1}^n b_r(j) v_j.$$

The cost becomes order nR . Sparse attention replaces the empirical measure by a restricted measure

$$\mu_i = \sum_{j \in S(i)} a_{ij} \delta_j,$$

where $S(i)$ is a local or learned support set. Both low-rank and sparse attention are finite approximations of the same normalized kernel operator.

5.8. Operators between high-dimensional function spaces

Neural operators learn maps

$$\mathcal{G} : \mathcal{A} \rightarrow \mathcal{V}$$

between function spaces. For example, in an elliptic PDE,

$$-\nabla \cdot (a(x) \nabla u(x)) = f(x), \quad x \in D \subset \mathbb{R}^r,$$

the solution operator is

$$\mathcal{G} : a \mapsto u.$$

The input a and output u are fields over D . A finite representation uses a mesh

$$D_h = \{x_1, \dots, x_n\}$$

and weights w_i , giving

$$a_h = \sum_{i=1}^n a(x_i) \ell_i, \quad u_h = \sum_{i=1}^n u(x_i) \ell_i$$

for basis functions ℓ_i .

A neural operator layer may be written as

$$v_{l+1}(x) = \sigma \left(W_l v_l(x) + \int_D K_l(x, y, a, v_l) v_l(y) dy \right).$$

A Fourier neural operator uses

$$v_{l+1}(x) = \sigma \left(W_l v_l(x) + \mathcal{F}^{-1} [M_l(\xi) \hat{v}_l(\xi)](x) \right),$$

with finitely many retained modes. A DeepONet-type representation has

$$\mathcal{G}(a)(y) \approx \sum_{r=1}^R b_r(a(x_1), \dots, a(x_m)) \tau_r(y),$$

where $x_1, \dots, x_m$ are sensors and τ_r are trunk basis functions.

The high-dimensional extension is governed by three approximation levels:

$$\mathcal{G} \longrightarrow \mathcal{G}_\theta \longrightarrow R_h \mathcal{G}_\theta P_h \longrightarrow \hat{\mathcal{G}}_{\theta, h}.$$

Here $\mathcal{G}_\theta$ is the parameterized operator class, P_h and R_h are input and output projections, and $\hat{\mathcal{G}}_{\theta, h}$ is the computable numerical implementation.

A resolution-stable neural operator should satisfy

$$\|\mathcal{G}(a) - R_h \mathcal{G}_\theta P_h a\|_{L^r} \rightarrow 0$$

as the representation is refined, uniformly for a in a specified compact or bounded subset of $\mathcal{A}$.

5.9. Compositional structure and dimension control

High-dimensional spaces require structural assumptions. Without structure, generic approximation rates deteriorate rapidly with dimension. The operator formulation makes the structural assumptions explicit through the kernel class.

A sparse interaction kernel has

$$K(i, j) = 0 \quad \text{for most pairs } (i, j).$$

A local geometric kernel has

$$K(x, y) = 0 \quad \text{when } d_X(x, y) > R.$$

A low-rank kernel has

$$K(x, y) = \sum_{r=1}^R a_r(x) b_r(y).$$

A multiscale kernel has

$$K(x, y) = \sum_{\lambda \in \Lambda} K_\lambda(x, y),$$

where K_λ acts at scale λ . A compositional operator has the form

$$\mathcal{T} = \mathcal{T}_L \circ \dots \circ \mathcal{T}_1,$$

where each $\mathcal{T}_l$ acts on a small subset of variables, a local neighborhood, a low-dimensional projection, or a restricted frequency band.

These structures explain why high-dimensional neural computation can be tractable. The learned map is rarely an arbitrary operator on all coordinates. It is built from local, sparse, equivariant, low-rank, multi-scale, or compositional operators.

5.10. High-dimensional stability

Let $\mathcal{T} : \mathcal{F}(X; E) \rightarrow \mathcal{G}(Y; F)$ be an operator approximated by $\mathcal{T}_h$. Stability requires a topology. For high-dimensional inputs, common choices include

$$L^p, \quad H^s, \quad B_{p,q}^s, \quad W_p, \quad \text{TV}, \quad \text{MMD}_k.$$

The correct topology depends on the data type. Images often use L^2 , Sobolev, Besov, or perceptual feature metrics. Point clouds often use Wasserstein or Hausdorff-type metrics. Distributional inputs often use MMD or Wasserstein distances. Graphs may use cut distance, spectral distance, or message-passing stability metrics.

Proposition 5.2 (Lipschitz stability of a nonlinear kernel layer). *Let*

$$(\mathcal{T}u)(x) = \Gamma \left(x, u(x), \int_X K(x, y) \Psi(u(y)) d\mu(y) \right).$$

Assume $\mu(X) < \infty$, $K \in L^\infty(X \times X; \mathcal{L}(G, H))$, $\Psi : E \rightarrow G$ is L_Ψ -Lipschitz, and $\Gamma : X \times E \times H \rightarrow F$ is Lipschitz in its second and third arguments with constants L_2, L_3 . Then

$$\|\mathcal{T}u - \mathcal{T}v\|_{L^\infty(X; F)} \leq (L_2 + L_3 \mu(X)) \|K\|_{L^\infty} L_\Psi \|u - v\|_{L^\infty(X; E)}.$$

Proof. For almost every x ,

$$\begin{aligned} \|\mathcal{T}u(x) - \mathcal{T}v(x)\| &\leq L_2 \|u(x) - v(x)\| \\ &\quad + L_3 \left\| \int_X K(x, y) (\Psi(u(y)) - \Psi(v(y))) d\mu(y) \right\| \\ &\leq L_2 \|u - v\|_{L^\infty} + L_3 \mu(X) \|K\|_{L^\infty} L_\Psi \|u - v\|_{L^\infty}. \end{aligned}$$

Taking the essential supremum over x proves the result. $\square$

This bound applies in any dimension. Dimension enters through the measure, the kernel norm, the regularity of the representation, and the feasibility of approximating the integral.

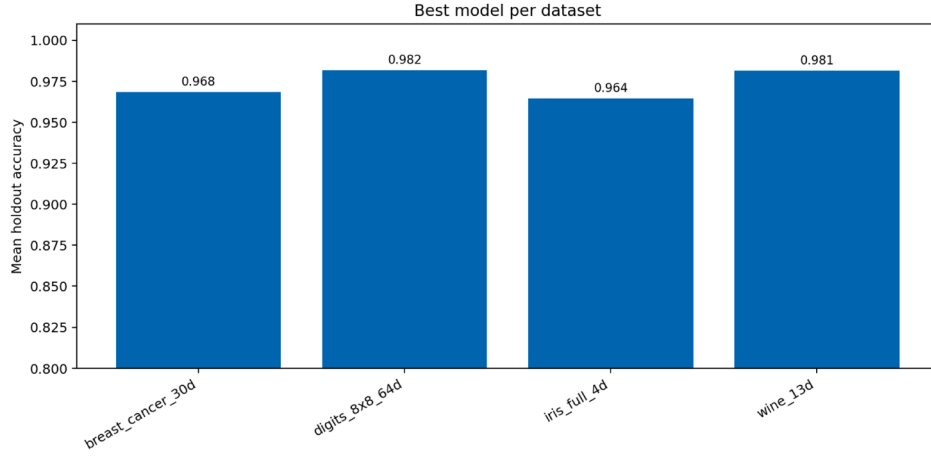


Fig. 1. Best repeated-holdout accuracy on each dataset. The figure summarizes the leading entry for each dataset in Table 1.

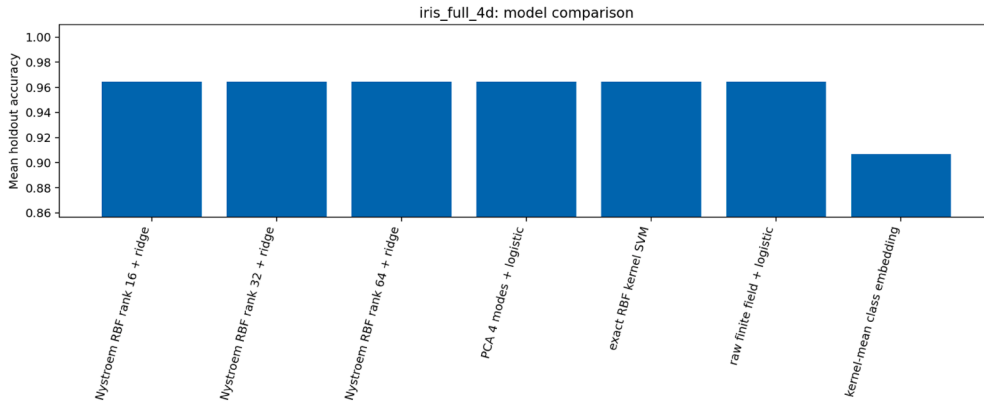


Fig. 2. Model comparison on the full four-dimensional Iris coordinate field. Several finite representations reach the same mean accuracy, showing that the task is nearly saturated by the raw coordinate field.

5.11. Empirical design for high-dimensional validation

A high-dimensional empirical study should vary the representation while holding the learning task fixed. For a dataset with samples u_i , labels y_i , and representation map P_m , train models

$$f_{\theta,m}(u) = g_{\theta}(P_m u)$$

for increasing m . Report

$$\text{Acc}(m), \quad \text{F1}(m), \quad \text{Cost}(m), \quad \text{Memory}(m).$$

For kernel methods, vary rank R :

$$k_R(x, y) = \sum_{r=1}^R \phi_r(x) \phi_r(y).$$

For graph methods, vary the number of nodes, graph sparsity, and spectral truncation. For point clouds, vary the number of sampled points and the bandwidth of the geometric kernel. For neural operators, vary the mesh resolution, number of Fourier modes, number of sensors, and output basis size.

A minimal high-dimensional experiment set should include:

tabular	$X = \{1, \dots, d\}$	dense, sparse, RBF, low-rank kernel
image	$X = \{1, \dots, H\} \times \{1, \dots, W\}$	CNN, PCA, Fourier, wavelet, kernel
video	$X = \{1, \dots, T\} \times \Omega_h$	spatiotemporal convolution, separable kernels
point cloud	$\mu_n = n^{-1} \sum_i \delta_{x_i}$	Deep Sets, attention, kernel mean
graph	$X = V(G)$	message passing, graph spectral filters
PDE field	$X = D \subset \mathbb{R}^r$	FNO, DeepONet, graph neural operator.

The central comparison is the tradeoff between approximation quality and computational structure. Dense kernels have maximal flexibil-

ity and high cost. Local kernels exploit geometry. Spectral kernels exploit smoothness. Low-rank kernels exploit compressibility. Sparse kernels exploit limited interactions. Multiscale kernels exploit scale separation. High-dimensional neural architecture design is the choice of the operator structure that gives the best approximation for the available representation budget.

6. Experiments

We evaluate the operator-theoretic formulation on finite coordinate fields, image fields, empirical kernel representations, and low-rank kernel approximations. The experiments use four standard classification datasets. Iris is represented as a full four-dimensional coordinate field. Wine is represented as a thirteen-dimensional coordinate field. Breast Cancer is represented as a thirty-dimensional coordinate field. Digits is represented as an 8×8 image field, equivalently a scalar function on a grid with 64 sites.

The purpose of these experiments is diagnostic and illustrative. They are designed to show how several familiar finite-dimensional procedures instantiate different representation and discretization choices within the same operator-theoretic formulation. They do not introduce a new predictive architecture, claim state-of-the-art performance, or empirically establish a universal approximation property. Instead, the experiments hold the underlying learning task fixed while varying projection dimension, harmonic truncation, or kernel rank, thereby making the representation-accuracy and representation-cost tradeoffs visible in concrete examples.

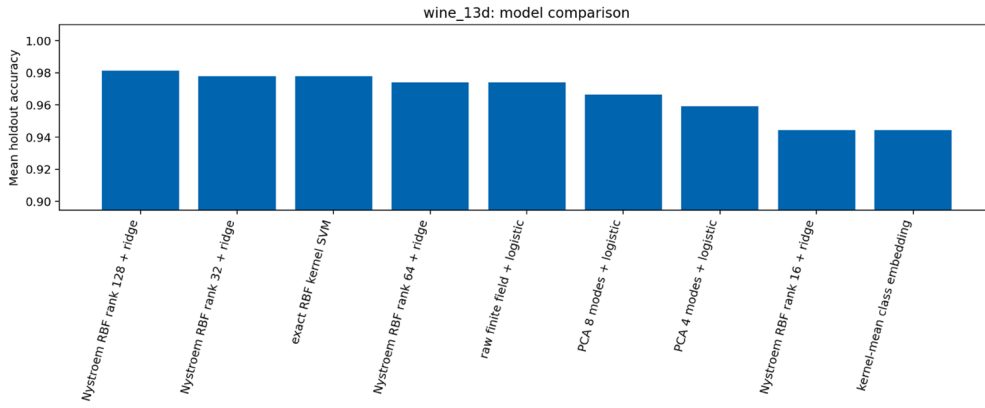


Fig. 3. Model comparison on the thirteen-dimensional Wine coordinate field. Nyström rank 128, exact RBF SVM, and raw finite-field logistic regression are the strongest models.

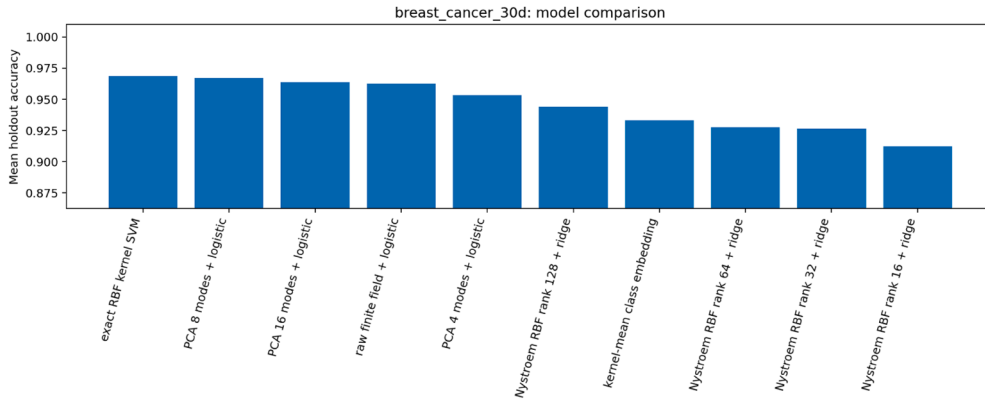


Fig. 4. Model comparison on the thirty-dimensional breast cancer coordinate field. The exact RBF SVM and PCA-truncated logistic models are strongest.

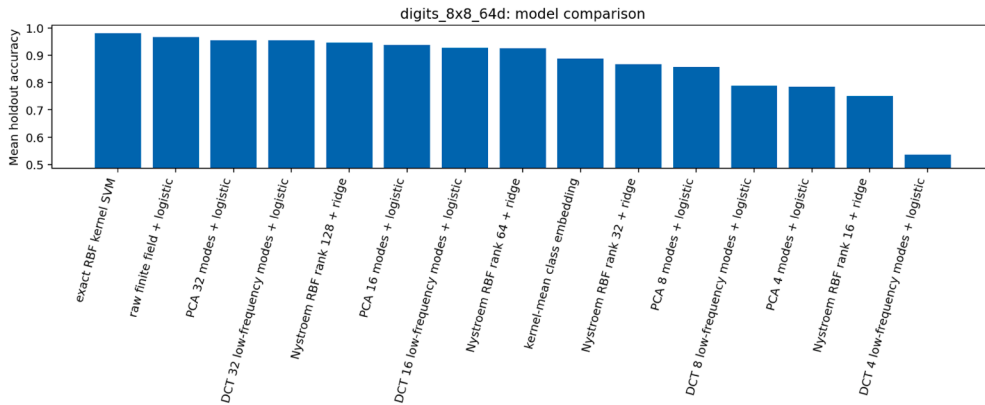


Fig. 5. Model comparison on the 8×8 digits image field. Full empirical kernel evaluation gives the highest accuracy, while raw grid values, PCA modes, DCT modes, and Nyström features provide progressively cheaper finite approximations.

Each experiment uses five stratified holdout splits with a 70/30 train-test split. Reported values are means and standard deviations across the five splits. The task metrics are accuracy and macro F1. The timing columns report mean fitting and prediction time in seconds.

The tested models correspond to different finite representations of the same underlying operator viewpoint. The raw logistic model acts directly on the finite coordinate or grid representation. PCA models use an empirical spectral projection before classification. DCT models use low-frequency harmonic coordinates on the image grid. The empirical kernel-mean classifier represents each class by a finite kernel mean embedding. The Nyström models approximate the RBF kernel by a finite

low-rank feature map. The exact RBF SVM evaluates the full empirical kernel operator over the training measure.

Table 1 reports the main repeated-holdout results. Fig. 1 summarizes the best model on each dataset. Dataset-specific model comparisons are shown in Figs. 2–5. The representation sweeps are reported in Table 2 and plotted in Figs. 6–9. The digits confusion matrix in Fig. 10 gives the class-level error structure for the strongest image-field model. Fig. 11 shows how the finite image field is reconstructed as the number of retained PCA modes increases.

The tabular results in Table 1 show that the finite-coordinate representation is already strong on Iris, Wine, and Breast Cancer. On Iris, raw logistic regression, PCA, Nyström, and the exact RBF SVM all reach

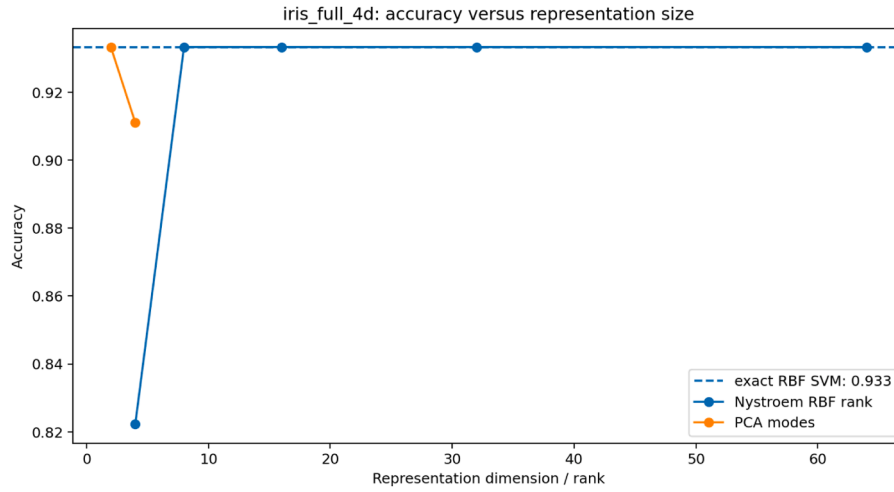


Fig. 6. Representation sweep on the Iris coordinate field. Rank 8 Nyström features already match the exact RBF baseline on the fixed split.

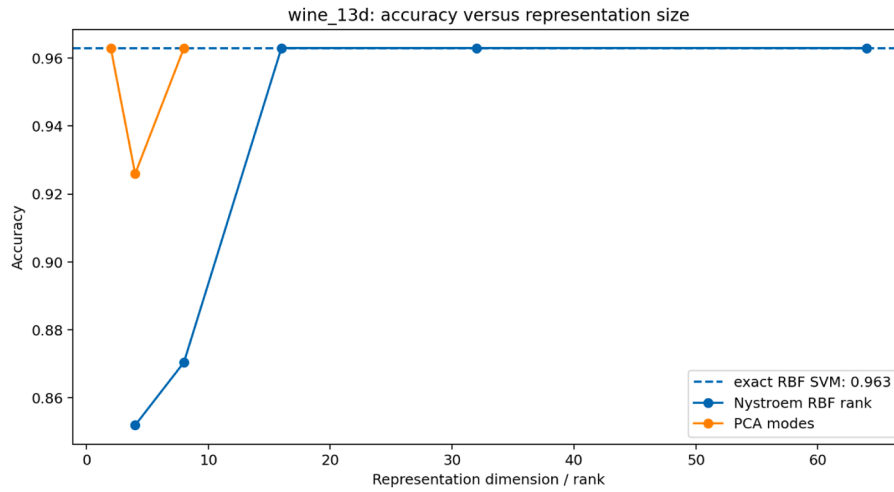


Fig. 7. Representation sweep on the Wine coordinate field. PCA and Nyström approximations both reach the exact RBF baseline at moderate dimension.

0.9644 mean accuracy. The class-kernel-mean classifier is weaker at 0.9067, indicating that the simple class-mean embedding loses discriminative information available to the supervised models. On Wine, the best model is the Nyström rank 128 approximation with 0.9815 accuracy, slightly ahead of the exact RBF SVM and the raw finite-field model. On Breast Cancer, the exact RBF SVM gives the best mean accuracy, 0.9684, with PCA 8 modes close behind at 0.9673. The PCA result indicates that much of the task-relevant variation is captured by a low-dimensional empirical spectral subspace.

The digits results are more representation-sensitive. The exact RBF SVM reaches 0.9819 mean accuracy, while raw logistic regression on the 64-site image field reaches 0.9667. PCA 32 and DCT 32 both reach approximately 0.956, showing that a compact projected field preserves most of the classification signal. The Nyström rank 128 model reaches 0.9474, and the kernel-mean class embedding reaches 0.8889. These results support the view that full empirical kernel evaluation, low-rank kernel approximation, and basis truncation are distinct finite approximations to operators acting on the same image field.

The representation sweeps in Table 2 make the approximation structure visible. On Breast Cancer, PCA improves from 0.9474 accuracy with two modes to 0.9708 with eight modes. Additional PCA modes do not improve the fixed-split result, suggesting that the discriminative information is concentrated in a small empirical spectral subspace. The Nyström approximation on the same dataset improves from rank 16 to rank 128, then remains below the exact RBF SVM.

The digits sweep gives the clearest high-dimensional field experiment. PCA rises from 0.7870 at four modes to 0.9741 at forty-eight modes. DCT rises from 0.5556 at four modes to 0.9741 at forty-eight modes. Nyström RBF features rise from 0.7833 at rank 16 to 0.9704 at rank 256. The exact RBF SVM reaches 0.9852 on the same split. This is the expected pattern when a full empirical kernel operator is approximated by finite-rank features and when an image field is approximated by truncated bases.

These experiments show that ordinary finite-dimensional datasets can be read as coordinate fields over finite measure spaces, and dense models, PCA models, kernel methods, and low-rank kernel approximations fit the same operator template. Further, image fields expose the approximation role of basis truncation: PCA and DCT accuracy increase as the retained representation becomes richer. Empirical kernel methods also give a direct finite-measure implementation of kernel operators, while Nyström features give a controlled finite-rank approximation whose performance improves with rank.

Appendix A repeats the image-field study on the full Digits dataset with all 1,797 samples. The appendix includes repeated-holdout results, representation sweeps, confusion matrices, and PCA/DCT reconstruction figures. Its main finding is that PCA, DCT, and Nyström approximations approach the raw-field and full-kernel baselines as the representation dimension or rank increases. This supports the interpretation of basis truncation and low-rank kernel maps as finite-dimensional approximations of operators acting on the same underlying image field.

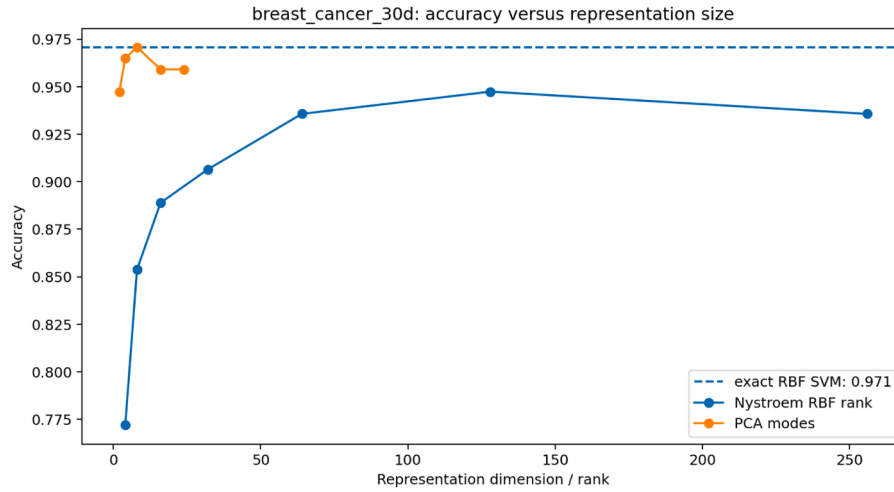


Fig. 8. Representation sweep on the Breast Cancer coordinate field. PCA reaches the strongest result at eight modes, while Nyström accuracy improves with rank.

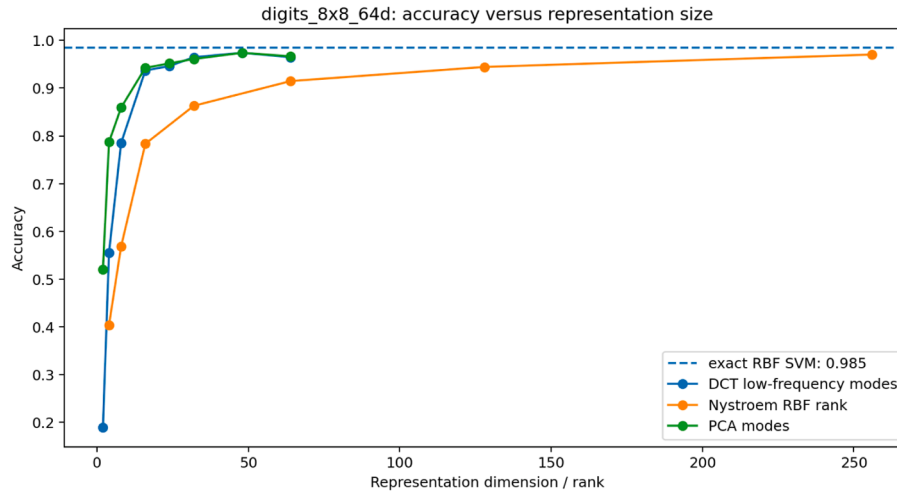


Fig. 9. Representation sweep on the Digits image field. PCA, DCT, and Nyström approximations improve as dimension or rank increases, with the exact RBF SVM providing the full-kernel reference.

Table 1

Repeated holdout results. Accuracy and macro F1 are averaged across five stratified 70/30 splits.

Dataset	Model	Accuracy	Macro F1	Fit time	Predict time
Iris 4d	raw finite field + logistic	0.9644 ± 0.0253	0.9643 ± 0.0254	0.0603	0.0020
Iris 4d	PCA 4 modes + logistic	0.9644 ± 0.0253	0.9643 ± 0.0254	0.0651	0.0045
Iris 4d	Nyström rank 16 + ridge	0.9644 ± 0.0253	0.9643 ± 0.0254	0.0224	0.0067
Iris 4d	exact RBF kernel SVM	0.9644 ± 0.0199	0.9644 ± 0.0199	0.0105	0.0044
Iris 4d	kernel-mean class embedding	0.9067 ± 0.0330	0.9058 ± 0.0337	0.0052	0.0060
Wine 13d	raw finite field + logistic	0.9741 ± 0.0166	0.9738 ± 0.0168	0.0473	0.0027
Wine 13d	PCA 8 modes + logistic	0.9667 ± 0.0155	0.9671 ± 0.0156	0.0303	0.0029
Wine 13d	Nyström rank 128 + ridge	0.9815 ± 0.0131	0.9813 ± 0.0135	0.1096	0.0131
Wine 13d	exact RBF kernel SVM	0.9778 ± 0.0203	0.9775 ± 0.0205	0.0123	0.0036
Wine 13d	kernel-mean class embedding	0.9444 ± 0.0293	0.9452 ± 0.0294	0.0029	0.0035
Breast Cancer 30d	raw finite field + logistic	0.9626 ± 0.0188	0.9598 ± 0.0201	0.0631	0.0019
Breast Cancer 30d	PCA 8 modes + logistic	0.9673 ± 0.0158	0.9650 ± 0.0168	0.0327	0.0037
Breast Cancer 30d	Nyström rank 128 + ridge	0.9439 ± 0.0158	0.9409 ± 0.0165	0.1202	0.0129
Breast Cancer 30d	exact RBF kernel SVM	0.9684 ± 0.0121	0.9664 ± 0.0129	0.0161	0.0060
Breast Cancer 30d	kernel-mean class embedding	0.9333 ± 0.0205	0.9264 ± 0.0235	0.0026	0.0102
Digits 8 × 8	raw finite field + logistic	0.9667 ± 0.0067	0.9667 ± 0.0066	0.4432	0.0032
Digits 8 × 8	PCA 32 modes + logistic	0.9563 ± 0.0031	0.9563 ± 0.0030	0.2132	0.0099
Digits 8 × 8	DCT 32 modes + logistic	0.9559 ± 0.0073	0.9558 ± 0.0073	0.2375	0.0464
Digits 8 × 8	Nyström rank 128 + ridge	0.9474 ± 0.0143	0.9474 ± 0.0142	0.0772	0.0144
Digits 8 × 8	exact RBF kernel SVM	0.9819 ± 0.0024	0.9818 ± 0.0024	0.1280	0.0585
Digits 8 × 8	kernel-mean class embedding	0.8889 ± 0.0077	0.8891 ± 0.0080	0.0034	0.0598

Table 2

Representation sweeps. Each row is evaluated on a fixed stratified 70/30 split. The exact RBF SVM is included as a full-kernel baseline.

Dataset	Representation	Dimension/rank	Accuracy	Macro F1
Iris 4d	PCA modes	2	0.9333	0.9333
Iris 4d	PCA modes	4	0.9111	0.9107
Iris 4d	Nyström RBF rank	4	0.8222	0.8214
Iris 4d	Nyström RBF rank	8	0.9333	0.9333
Iris 4d	exact RBF SVM	–	0.9333	0.9327
Wine 13d	PCA modes	2	0.9630	0.9636
Wine 13d	PCA modes	4	0.9259	0.9285
Wine 13d	PCA modes	8	0.9630	0.9619
Wine 13d	Nyström RBF rank	4	0.8519	0.8510
Wine 13d	Nyström RBF rank	16	0.9630	0.9619
Wine 13d	exact RBF SVM	–	0.9630	0.9619
Breast Cancer 30d	PCA modes	2	0.9474	0.9436
Breast Cancer 30d	PCA modes	4	0.9649	0.9625
Breast Cancer 30d	PCA modes	8	0.9708	0.9687
Breast Cancer 30d	PCA modes	16	0.9591	0.9562
Breast Cancer 30d	Nyström RBF rank	16	0.8889	0.8849
Breast Cancer 30d	Nyström RBF rank	64	0.9357	0.9323
Breast Cancer 30d	Nyström RBF rank	128	0.9474	0.9443
Breast Cancer 30d	exact RBF SVM	–	0.9708	0.9687
Digits 8 × 8	PCA modes	4	0.7870	0.7855
Digits 8 × 8	PCA modes	16	0.9426	0.9431
Digits 8 × 8	PCA modes	32	0.9611	0.9611
Digits 8 × 8	PCA modes	48	0.9741	0.9741
Digits 8 × 8	DCT modes	4	0.5556	0.5422
Digits 8 × 8	DCT modes	16	0.9370	0.9376
Digits 8 × 8	DCT modes	32	0.9648	0.9648
Digits 8 × 8	DCT modes	48	0.9741	0.9740
Digits 8 × 8	Nyström RBF rank	16	0.7833	0.7762
Digits 8 × 8	Nyström RBF rank	64	0.9148	0.9142
Digits 8 × 8	Nyström RBF rank	128	0.9444	0.9445
Digits 8 × 8	Nyström RBF rank	256	0.9704	0.9702
Digits 8 × 8	exact RBF SVM	–	0.9852	0.9852

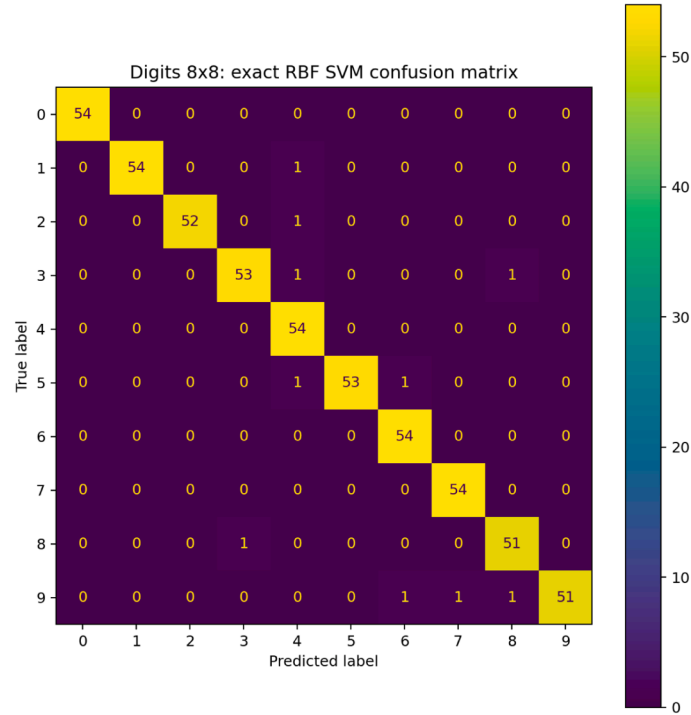


Fig. 10. Confusion matrix for the exact RBF SVM on the Digits image field. The small off-diagonal counts show that the full empirical kernel operator separates most digit classes on the fixed split.

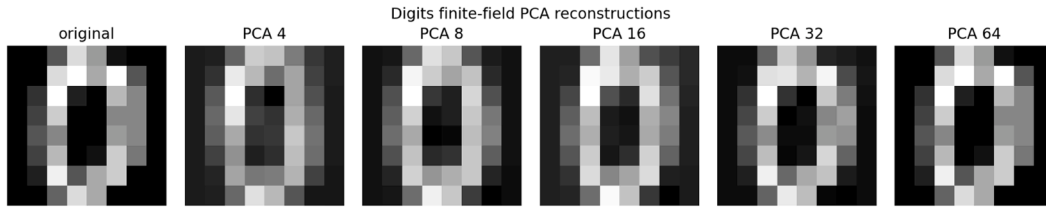


Fig. 11. PCA reconstructions of a Digits image field. Increasing the number of retained modes improves the finite representation of the underlying grid function.

7. Discussion

The preceding sections present neural architectures as finite-dimensional computations of operators on functions and measures. The array manipulated by a network is often a representation of a more structured object. A vector can be a function on a finite coordinate set. An image can be a sampled field. A point cloud can be an empirical measure. A graph signal can be a function on vertices with interactions determined by edges. A token sequence can be a feature-bearing empirical measure with a state-dependent Markov kernel. With this distinction in place, dense layers, convolutions, graph message passing, attention, neural fields, kernel methods, and neural operators can be treated as different finite evaluations of kernel-based operators.

This perspective clarifies the role of architecture. A neural architecture specifies more than a parameterized map between arrays. It specifies a domain, measure, kernel structure, representation, topology, and numerical approximation rule. Dense networks use finite sets with unrestricted kernels. Convolutional networks impose translation structure. Graph networks restrict interactions through adjacency, diffusion, or graph spectra. Attention uses a normalized state-dependent kernel over an empirical measure. Neural operators learn maps between function spaces while aiming to retain meaning across discretizations. These choices affect approximation, stability, computational cost, and generalization.

The experiments illustrate this point in finite dimensions. Tabular datasets can be read as coordinate fields over finite measure spaces. Image datasets can be read as sampled scalar fields over grids. PCA and DCT models are projection methods: they replace the original field by basis coefficients. Nyström models are finite-rank approximations of kernel operators. Empirical kernel-mean classifiers represent classes by finite kernel embeddings. Exact RBF SVMs evaluate full empirical kernel operators. The observed sweeps show the expected approximation behavior: increasing the number of retained modes or the kernel rank generally improves performance, while exact empirical kernel evaluation gives a strong full-kernel reference. The full Digits experiment in [Appendix A](#) gives the clearest example, since the 8×8 image field admits raw-grid, PCA, DCT, low-rank kernel, and full-kernel representations of the same learning problem.

The results also show why no single representation is uniformly best. A representation is effective when it matches the structure of the target operator and the topology used to measure error. Fourier and DCT representations suit smooth or globally oscillatory structure. Wavelets suit localized multiscale variation. Local kernels and convolutions exploit spatial locality. Low-rank kernels exploit compressibility. Graph spectral filters exploit diffusion geometry. Empirical measure methods suit sets, point clouds, bags, and distributions. Attention is effective when interactions should be data-dependent and normalized over a finite support. The same input can support several valid finite representations, and the best choice depends on the regularity, invariance, and interaction pattern of the operator being learned.

The operator view also sharpens the meaning of stability. A model is stable only relative to a specified topology. Weak convergence is appropriate for bounded continuous test functions on measures. Wasserstein distance is appropriate when the geometry of the domain matters. Total

variation is appropriate when mass changes must be controlled strongly. Sobolev and Besov norms are appropriate for fields where derivatives or multiscale regularity matter. RKHS metrics are appropriate when distributions are compared through kernel evaluations. Without specifying the topology, claims about robustness, discretization invariance, or resolution transfer are underspecified.

This formulation suggests concrete reporting standards for experiments. Accuracy alone is not enough. One should also report the finite representation used to obtain it: grid size, number of points, number of retained modes, graph size, kernel rank, attention sparsity, quadrature rule, or mesh resolution. For neural operators and field-based models, performance should be measured under resolution change. For low-rank kernels and efficient attention, performance should be plotted against rank. For basis methods, performance should be plotted against the number of retained coefficients. For empirical measures, performance should be plotted against sample size and sampling rule. These curves indicate whether a model is approximating a stable underlying operator or fitting a particular discretization.

There are limitations to the present treatment. The paper gives a broad formulation rather than sharp optimal rates for each architecture class. The stability estimates are general; sharper results require assumptions on smoothness, geometry, kernel regularity, sampling schemes, and optimization. The experiments are diagnostic and small-scale. Their role is to verify that the proposed distinctions between the represented object, the abstract operator, and its finite computational realization can be identified in standard learning procedures. They should therefore be interpreted as demonstrations of the framework rather than as evidence for a new architecture, a general approximation rate, or a universal performance advantage. They demonstrate the representation and approximation viewpoint, but they do not replace large benchmarks for images, graphs, language, point clouds, or PDE solution operators. The treatment of training dynamics is also limited. Optimization, implicit regularization, and finite-sample generalization interact with representation error and operator approximation error, and a complete account must study these interactions together.

Several directions follow. One direction is to prove architecture-specific approximation theorems that decompose error into projection error, quadrature error, kernel-class error, optimization error, and statistical error. Another is to study discretization-invariant learning for graph neural operators, Fourier neural operators, and attention models on variable-size empirical measures. A third is to design adaptive representations that change the sites, weights, basis functions, or kernel ranks during learning. A fourth is to build benchmarks that vary representation size, sampling density, graph resolution, or spectral bandwidth while holding the underlying task fixed. Such benchmarks would test whether a learned model approximates an operator rather than a fixed array map.

The main conclusion is that many neural computations can be understood as numerical evaluations of structured operators through finite representations. This does not replace the usual finite-dimensional view of neural networks, but it does explain how that view fits into a broader approximation problem. Matrices, convolutions, attention weights, graph adjacencies, Fourier multipliers, kernel features, and empirical measures are all ways of representing and applying operators.

Treating them as such gives a principled way to compare architectures, reason about stability, and design models suited to the structure of the data and the operator being learned.

CRedit authorship contribution statement

Míngshū Wáng: Writing – review & editing, Writing – original draft, Visualization, Validation, Methodology, Investigation, Formal analysis, Data curation, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgement

I would like to thank EIGENFORM for their support and help in providing the literature, especially CEO Jennifer Dodgson for her essays on neural computing.

Appendix A. Full digits image-field experiment

The full Digits experiment uses all 1,797 images in the scikit-learn handwritten digits dataset. Each image is an 8×8 grayscale field, represented as a function

$$u : \Omega_h \rightarrow \mathbb{R}, \quad \Omega_h = \{1, \dots, 8\} \times \{1, \dots, 8\}.$$

Equivalently, each sample is a finite scalar field with 64 grid sites. The task is ten-class classification.

The experiment compares six finite representations or operator approximations. The raw logistic model acts directly on the 64-site image field. PCA models project the image field onto an empirical spectral basis before classification. DCT models project the image field onto fixed low-frequency harmonic coordinates. The empirical kernel-mean classifier represents each digit class by a finite RBF kernel mean embedding. The Nyström models approximate the RBF kernel operator by finite-rank features. The exact RBF SVM evaluates the full empirical kernel operator on the training measure.

Repeated-holdout results use five stratified 70/30 train-test splits. Table A.1 reports mean accuracy, mean macro F1, and timing across

the five splits. Fig. A.1 plots the same model comparison. The exact RBF SVM gives the strongest repeated-holdout performance, with mean accuracy 0.9819 and mean macro F1 0.9818. Raw logistic regression on the full 64-pixel field reaches 0.9667 accuracy. PCA with all 64 modes gives the same accuracy as raw logistic regression, as expected because the representation preserves the full input up to an invertible linear transform after standardization. PCA 48, DCT 48, and Nyström rank 256 are all within two percentage points of the exact RBF SVM while using compressed or finite-rank representations.

A second experiment fixes one stratified 70/30 split and varies the representation dimension or kernel rank. Table A.2 reports this sweep, and Fig. A.2 plots the accuracy curves. PCA accuracy rises from 0.5204 at two modes to 0.9741 at forty-eight modes. DCT accuracy rises from 0.1889 at two low-frequency modes to 0.9741 at forty-eight modes. Nyström accuracy rises from 0.4037 at rank 4 to 0.9759 at rank 512. The exact RBF SVM reaches 0.9852 on the same split.

The sweep in Table A.2 makes the approximation role of the representation explicit. PCA and DCT are projections of the image field onto finite bases. Nyström features are a finite-rank approximation of the RBF kernel operator. Increasing modes or rank reduces representation error until the remaining gap is mainly due to the model class and classifier. The nonmonotonic values at full dimension, such as PCA 64 being slightly below PCA 48 on the fixed split, reflect finite-sample fitting and regularization rather than failure of the representation itself.

Fig. A.3 shows the confusion matrix for the exact RBF SVM on the fixed split. The off-diagonal mass is small, consistent with the 0.9852 accuracy reported in Table A.2. Figs. A.4 and A.5 visualize the image-field reconstructions obtained from PCA and DCT truncations. The reconstruction figures support the same conclusion as the classification sweep: low-dimensional projections initially discard important stroke information, while moderate-dimensional projections preserve enough structure for high classification accuracy.

The full Digits experiment supports the operator-theoretic interpretation in a concrete high-dimensional finite setting. The raw pixel model treats each image as a finite field over a grid. PCA and DCT replace the grid field by finite basis coefficients. Nyström features approximate the empirical RBF kernel operator by a finite-rank map. The exact RBF SVM evaluates the full empirical kernel operator. The observed accuracy curves show the expected approximation pattern: richer projections and higher kernel ranks approach the full-field or full-kernel baselines, while compressed representations expose the cost-accuracy tradeoff induced by finite-dimensional operator approximation.

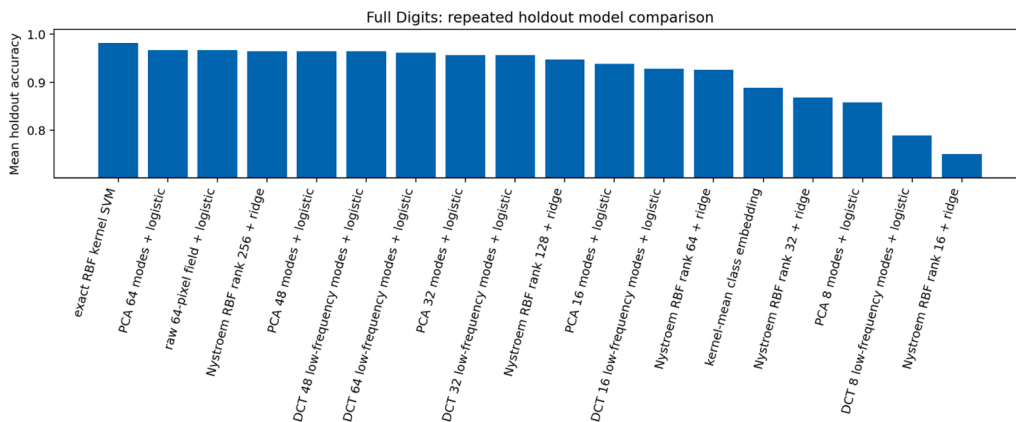


Fig. A.1. Full Digits model comparison using all 64 pixels. The exact RBF SVM is strongest, while PCA, DCT, and Nyström approximations approach the raw-field and full-kernel models as their representation dimension increases.

Table A.1

Full Digits repeated-holdout results over five stratified 70/30 splits. Column headings are split over two rows to save horizontal space.

Model	Accuracy mean	Accuracy sd	Macro F1 mean	Macro F1 sd	Fit sec.	Predict sec.
Exact RBF kernel SVM	0.9819	0.0024	0.9818	0.0024	0.1851	0.0773
PCA 64 modes + logistic	0.9667	0.0067	0.9667	0.0066	0.4914	0.0076
Raw 64-pixel field + logistic	0.9667	0.0067	0.9667	0.0066	0.8168	0.0041
Nyström RBF rank 256 + ridge	0.9648	0.0073	0.9649	0.0072	0.3333	0.0388
PCA 48 modes + logistic	0.9648	0.0056	0.9648	0.0055	0.4179	0.0070
DCT 48 low-frequency modes + logistic	0.9644	0.0084	0.9643	0.0083	0.4013	0.0714
DCT 64 low-frequency modes + logistic	0.9619	0.0061	0.9618	0.0060	0.4482	0.0704
PCA 32 modes + logistic	0.9563	0.0031	0.9563	0.0030	0.1954	0.0087
DCT 32 low-frequency modes + logistic	0.9559	0.0073	0.9558	0.0073	0.1680	0.0352
Nyström RBF rank 128 + ridge	0.9474	0.0143	0.9474	0.0142	0.1428	0.0287
PCA 16 modes + logistic	0.9378	0.0038	0.9377	0.0043	0.2400	0.0075
DCT 16 low-frequency modes + logistic	0.9278	0.0039	0.9274	0.0043	0.1231	0.0264
Nyström RBF rank 64 + ridge	0.9256	0.0187	0.9258	0.0181	0.0793	0.0122
Kernel-mean class embedding	0.8889	0.0077	0.8891	0.0080	0.0111	0.0693
Nyström RBF rank 32 + ridge	0.8681	0.0189	0.8684	0.0180	0.0362	0.0077
PCA 8 modes + logistic	0.8578	0.0056	0.8583	0.0058	0.3151	0.0045
DCT 8 low-frequency modes + logistic	0.7889	0.0139	0.7870	0.0143	0.1697	0.0254
Nyström RBF rank 16 + ridge	0.7507	0.0252	0.7426	0.0310	0.0253	0.0054

Table A.2

Full Digits representation sweep on a fixed stratified 70/30 split. The PCA and DCT rows vary projection dimension; the Nyström rows vary RBF kernel rank.

Representation	Dimension or rank	Accuracy	Macro F1
PCA modes	2	0.5204	0.5079
PCA modes	4	0.7870	0.7855
PCA modes	8	0.8593	0.8597
PCA modes	16	0.9426	0.9431
PCA modes	24	0.9519	0.9521
PCA modes	32	0.9611	0.9611
PCA modes	48	0.9741	0.9741
PCA modes	64	0.9667	0.9666
DCT low-frequency modes	2	0.1889	0.1539
DCT low-frequency modes	4	0.5556	0.5422
DCT low-frequency modes	8	0.7852	0.7826
DCT low-frequency modes	16	0.9370	0.9376
DCT low-frequency modes	24	0.9463	0.9465
DCT low-frequency modes	32	0.9648	0.9648
DCT low-frequency modes	48	0.9741	0.9740
DCT low-frequency modes	64	0.9648	0.9648
Nyström RBF rank	4	0.4037	0.3524
Nyström RBF rank	8	0.5685	0.5373
Nyström RBF rank	16	0.7833	0.7762
Nyström RBF rank	32	0.8630	0.8609
Nyström RBF rank	64	0.9148	0.9142
Nyström RBF rank	128	0.9444	0.9445
Nyström RBF rank	256	0.9704	0.9702
Nyström RBF rank	512	0.9759	0.9757
Exact RBF SVM baseline	–	0.9852	0.9852

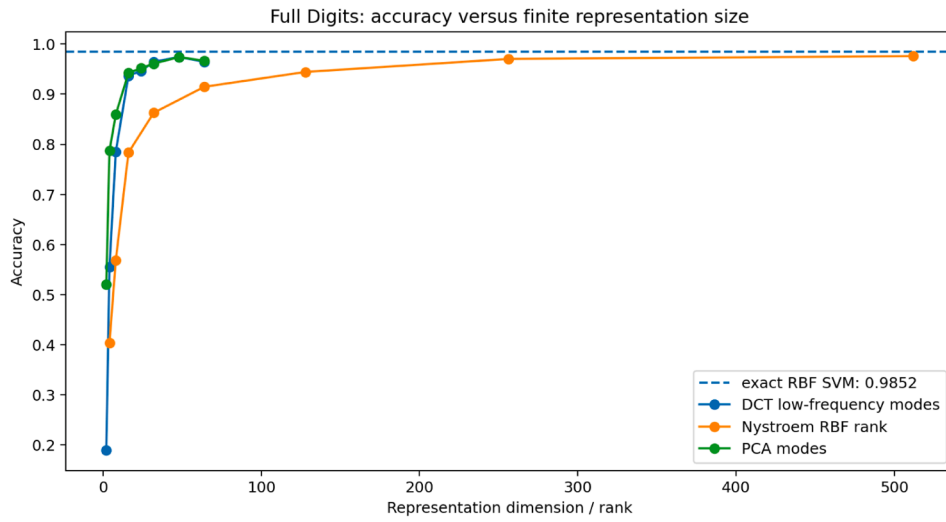


Fig. A.2. Full digits accuracy as a function of finite representation size. PCA modes, DCT modes, and Nyström ranks all improve as the representation becomes richer. The dashed line gives the exact RBF SVM reference.

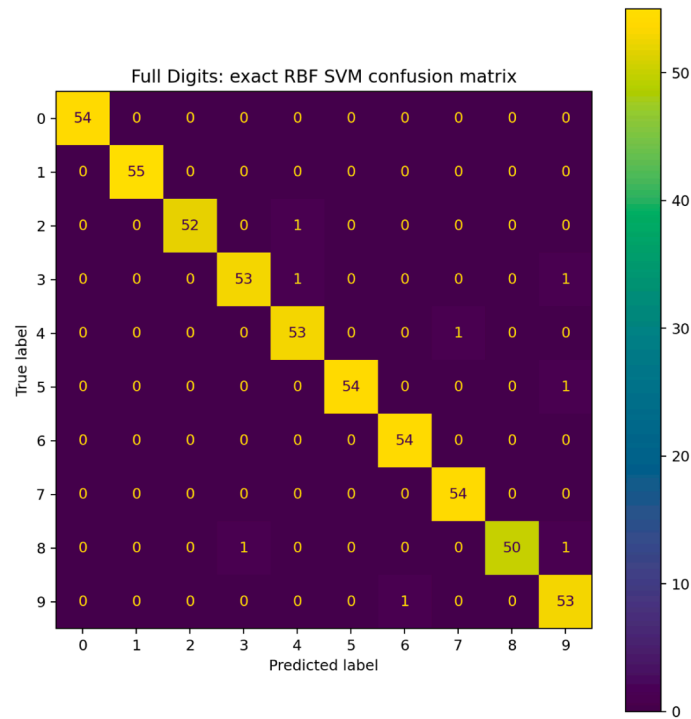


Fig. A.3. Confusion matrix for the exact RBF SVM on the fixed full-Digits split. The small off-diagonal counts show that the full empirical kernel operator separates most digit classes.

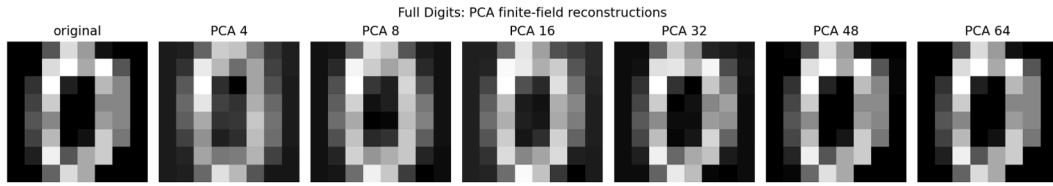


Fig. A.4. PCA reconstructions of a Digits image field. Increasing the number of retained empirical modes improves the finite representation of the 8×8 grid function.

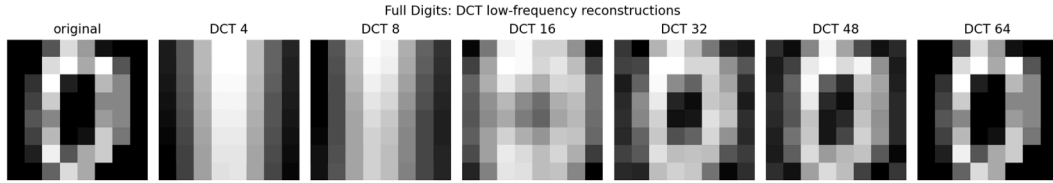


Fig. A.5. DCT low-frequency reconstructions of a Digits image field. The low-frequency harmonic representation becomes accurate once enough modes are retained.

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